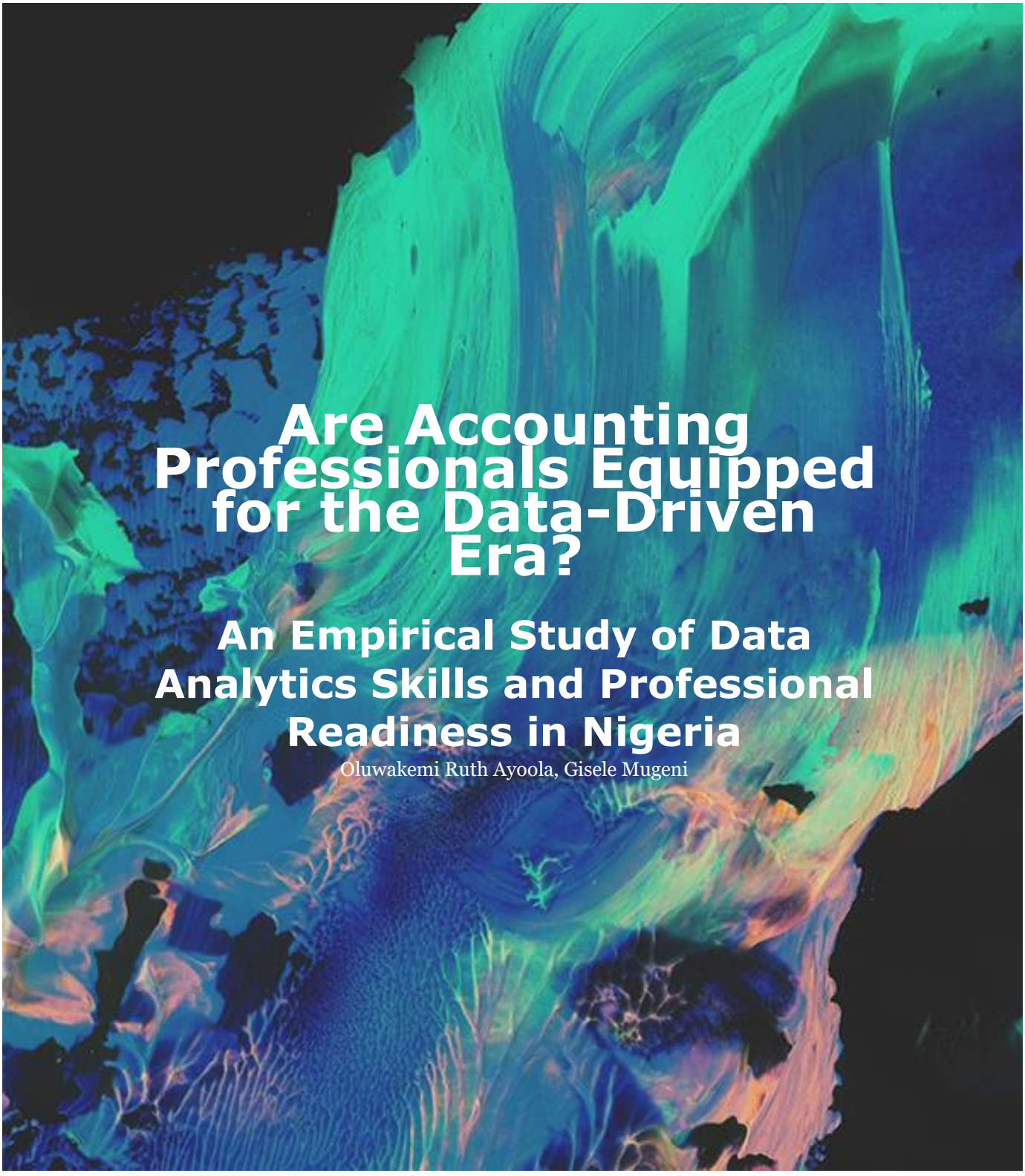
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Are Accounting Professionals Equipped for the Data-Driven Era?

An Empirical Study of Data Analytics Skills and Professional Readiness in Nigeria

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Abstract

Digitalisation has radically changed the demands of accounting professionals from firms operating today. All business organisations make decisions based on data analytics, which requires more from the accounting staff than simply preparing standard reports. Nevertheless, despite this shift in practices, there exists very limited information about the actual analytics competencies of accounting practitioners in emerging countries. Therefore, the purpose of this study is to examine the extent of preparedness of accounting professionals working in Nigeria for data analytics.

Professional competence theory (Mulder, 2014), human capital theory (Becker, 1993), and the role theory (Biddle, 1986) serve as frameworks guiding the study. According to professional competence theory, readiness is defined as an evaluation by professionals of their capacity to perform their duties. According to human capital theory, analytics abilities can be developed via education and training. The role theory highlights changes in institutional requirements and role conflicts because of lack of professional training.

For the research design, convergent mixed methods were used: an online survey of 125 Nigerian accountants and financial managers together with six semi structured interviews. Data collected via surveys were analysed using descriptive statistics and chi-square tests in jamovi. Data obtained from interviews were processed using Gioia methodology of coding.

The results reveal partial and conditional preparedness of Nigerian accountants for the data analytics era. Advanced Excel is the key tool used by accountants, whereas powerful platforms such as PowerBI and Tableau are used rarely. Although most respondents feel ready, the reality is more complex: perceived readiness appears to be conditional valid in case of familiar tasks but unreliable for new analytics challenges. Conditional readiness is introduced to describe this phenomenon. Learning on one's own prevails among the ways accountants develop analytics skills, although university education, although considered theoretically sufficient, is viewed as not enough in practice. Chi-square test shows that perceived educational preparedness, analytical self-confidence, and organisational support are the main predictors of conditional readiness. No significant association was revealed between years of experience, position, and qualification level on the one hand, and conditional readiness on the other.

This study contributes to the body of accounting knowledge by offering evidence on the concept of conditional readiness, using human capital theory and role theory in relation to emerging economies, and highlighting that being ready to deal with data analytics does not necessarily depend on seniority and qualifications but rather on the effectiveness of development of capabilities. Recommendations suggested include those aimed at universities, professional bodies (ICAN, ACCA), and employers.

Keywords: data analytics, professional readiness, accounting profession, Nigeria, human capital theory, role theory, professional competence, mixed methods, conditional readiness

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List of Abbreviations

ACCA	Association of Chartered Certified Accountants
AI	Artificial Intelligence
ANAN	Association of National Accountants of Nigeria
BI	Business Intelligence
CFI	Corporate Finance Institute
CPD	Continuing Professional Development
ERP	Enterprise Resource Planning
GDP	Gross Domestic Product
HND	Higher National Diploma
ICAN	Institute of Chartered Accountants of Nigeria
IFAC	International Federation of Accountants
IFRS	International Financial Reporting Standards
NASB	Nigerian Accounting Standards Board
OND	Ordinary National Diploma
SAS	Statement of Accounting Standards
SPSS	Statistical Package for the Social Sciences
USBE	Umeå School of Business, Economics and Statistics

1. Introduction

1.1 Background and Practical Motivation

The evolution of the last twenty years saw a profound change within the accounting profession. The rapid advance of digital technologies, ERP software, cloud computing, and business intelligence solutions resulted in a dramatic expansion of accounting tasks far beyond its conventional boundaries of record keeping, report writing, and compliance management (Burns and Baldvinsdottir, 2005, p. 726; Appelbaum et al., 2017, p. 30).

However, such change goes far beyond technological shifts. In essence, it represents the transformation of the very role that an accountant is supposed to perform within an organization as well as a shift in the scope of stakeholders who rely on the accountant's expertise. Accountants must process vast amounts of information including both financial and nonfinancial data; discover hidden patterns and emerging trends; and use this analytical output to derive meaningful recommendations for managers (Bhimani and Willcocks, 2014, p. 470).

The aforementioned global changes intersect with local realities of an emerging economy. Nigeria presents a relevant case of such a country, being one of the most powerful African economies by GDP. Its rapid development results in increasing organizational complexity and sophistication of processes requiring involvement of a competent analyst who can provide a comprehensive understanding of the situation in question and offer appropriate responses (Iyoha and Oyerinde, 2010, p. 362).

Yet, despite the need for such competencies, the situation within the professional field is characterized by some diversity related to the accessibility of analytic training and the level of analytics proficiency among accountants. In particular, many professionals acquire data analytics skills through self-study and practical experience while no formal training programs exist at the institutional level (Fijabi and Lasisi, 2023, p. 65-72). At the same time, neither ICAN nor ACCA have taken steps towards introducing advanced analytical skills into qualifications and CPD programs. University curriculums remain dominated by theoretical knowledge of accounting standards.

Therefore, analytics readiness in Nigeria cannot be attributed to lack of motivation or willingness to develop. Instead, it reflects an institutional mismatch between contemporary needs and the existing professional training system which becomes even more apparent when compared to the requirements that modern accounting roles place upon practitioners. Hence, the study seeks to investigate the nature, extent, and causes of the described mismatch.

1.2 Theoretical Background and Topicality

From the standpoint of Business Administration, the evolution of accounting work is an example of broader changes related to the alteration of professional roles and competencies. The evolution of the accountant's role from a simple recorder of transactions to a partner for business decisions has been discussed in many studies (Granlund and Lukka, 1998, p. 153; Burns and Scapens, 2000, p. 7). However, very little has been said about how this process is experienced by practitioners in countries with unique institutional conditions, like Nigeria.

Three theories will be used in the analysis. First, professional competence theory defines competence as a set of knowledge and skills together with their effective application in certain situations (Mulder, 2014, p. 111). Its key point is that competence includes the subjective perception of readiness for dealing with challenges posed by the profession's tasks (Mulder, 2014, p. 114). Thus, it is particularly relevant for this study, in which the mismatch between what professionals know and what they can do might be quite significant.

Secondly, human capital theory allows looking at knowledge and skills as assets that grow with learning and experience and increase productivity (Becker, 1993, p. 19-20). It enables exploring the dynamics of accounting professionals' acquisition of analytics capabilities and revealing the reasons behind the observed discrepancy in their readiness, even if the differences in the level of education are small. In particular, the theory stresses that the profitability of investment in human capital depends on how relevant skills become (Nafukho et al., 2004, p. 545-546). The point is very valid nowadays when conventional accounting skills might not ensure analytics readiness due to technological changes.

Thirdly, role theory helps understand how professional roles change along with alterations of the institutional and organisational setting (Biddle, 1986, p. 68). Concepts of role conflict and role ambiguity proposed by Biddle (1986, p. 82-83) appear to be directly applicable to accountants' situation as they are expected to contribute to analytics, while having insufficient skills due to inadequate educational background.

Thus, the above three theories cover three different aspects of the problem. Role theory reveals the conditions under which role expectations emerge. Human capital theory examines the mechanism behind capability development. Finally, professional competence theory describes the subjective side of the resulting readiness. Together, they allow developing a theoretical basis for the conceptual framework discussed in Section 2.7.

As mentioned earlier, the post-COVID period added urgency to this issue. Digital transformation has occurred suddenly. The use of remote working and digital technologies to manage finances and processes became essential almost overnight. Therefore, the importance of analytic skills among accountants grew considerably.

1.3 Research Gap and Positioning

There is existing academic literature that has made a meaningful contribution towards gaining an understanding of the impact of data analytics on the accounting profession. Literature in this field predominantly falls under one stream accounting education research on the incorporation of analytics abilities into the educational curriculum and the perceptions of accounting students and educators concerning the skills needed in future professional life (Richins et al., 2017, p. 67; Lawson et al., 2014, p. 295). This body of work has provided comprehensive knowledge regarding the increasing necessity of analytics skills among graduates and the concrete competencies that they include data visualisation and analysis ability, statistical analysis ability, and ability to use business intelligence tools.

However, education research provides information regarding future professional skills. It explains what students are trained for, but does not explain whether they will be able to successfully employ these skills in practice when becoming a professional accountant. In

Nigeria, the problem is acute because the majority of professionals received their education at a time when data analytics was not a matter of great importance yet, and because skill acquisition in Nigeria occurs via self-learning rather than education.

Research on professional skills and data analytics has been primarily done in North American and European countries. For example, Appelbaum et al. (2017, p. 41-42) and Sutton et al. (2016, p. 68) prove that there is always a gap between employers' expectations for data analytics abilities of professionals and those actual abilities. However, these studies fail to take into account the specifics of the professional accounting environment in Nigeria. Hence, there is a clear research gap since little research was done in this area in spite of the economic significance of Nigeria's accounting environment.

This study seeks to bridge both mentioned gaps. Differently from previous literature, the current study is not aimed at analysing accounting education and curriculum. The focus of this study is practice. It analyses the tools used by accountants in practice, the way they perceive their readiness for analytics tasks, and skill acquisition mechanisms. The study uses such theories as role theory, human capital theory, and professional competence theory. Moreover, the methodology chosen includes a convergent mixed method design with the use of survey data from 125 participants and depth interviews with six professionals.

1.4 Purpose of the Study

The purpose of this study is to assess the level at which accountants in Nigeria are adequately equipped with data analytics skills within a data-driven accounting setting. This involves identifying the tools and techniques used by practitioners, perceptions about their readiness, learning paths for acquiring analytics skills throughout their career, and the problems faced in mastering and using those skills. This research seeks to provide empirical data that will guide university programs, professional organizations, and corporations in bridging the gap between the expectation and actuality of analytics skills among accountants in Nigeria.

1.5 Research Questions

The above purpose gives rise to the following main research question:

'To what extent are accounting professionals working in Nigeria equipped with data analytics skills required in a data driven accounting environment?'

This question is addressed through four sub research questions:

- What data analytics tools and techniques are used by accounting professionals working in Nigeria in their professional roles?
- How do accounting professionals working in Nigeria perceive their readiness to perform analytics related tasks in practice?
- Through which pathways do accounting professionals working in Nigeria acquire data analytics skills during their professional careers?
- What challenges do accounting professionals working in Nigeria encounter when developing and applying data analytics skills in practice?

1.6 Scope and Delimitations

This study addresses professionals who practice accountancy and finance in Nigeria. The study is not about curriculum evaluation and comparison of educational practices, and therefore the discussion will be oriented around professional skills only. While the survey questionnaire was conducted using an electronic distribution method, it should be pointed out that some responses (7.2%) were provided by professionals employed in countries other than Nigeria. Despite the fact that non-Nigerian responses are not the key component of analysis, they will still be used to support the overall scheme of analysis and investigation.

Self-perceived readiness and skill usage are being measured and analysed in the present study. The reasons behind such an approach to measuring professional readiness should be found in the theory of professional competence. Specifically, it should be mentioned that according to Mulder (2014, p. 114), perceived readiness constitutes an important dimension of professional readiness. The issue of data biases inherent in self-reporting is discussed in Section 3.7 and 6.6. Audit firms are not included as the objects of study, and neither is statutory financial reporting nor the process of implementation of accounting standards in place.

1.7 Structure of the Thesis

Chapter 2 provides the theoretical framework and literature review that elaborates the theoretical background with a critique of theories of professional competence, human capital theory, role theory, and empirical literature on data analytics in accounting practice. Chapter 3 outlines the methodology, comprising the philosophical approach, research design, sampling method, research tools for data collection, measurement of constructs, analytical techniques, standards of research quality, and ethical considerations. Chapter 4 provides the results of the survey and the semi-structured interviews in terms of descriptive statistics, chi-square associations, qualitative Gioia analysis with the full dataset, and a combination of both types of data. In Chapter 5, the results are analyzed and discussed using the three theoretical perspectives, where conditional readiness is developed, as well as the theoretical contributions of the study. In Chapter 6, the conclusions of the study summarize the results, theoretical contributions, recommendations, ethical and social implications, limitations, and further directions for research.

1.8 Nigeria's Digital Transformation Context

Understanding the analytics readiness challenge requires appreciation of the broader digital transformation context in which Nigerian accounting professionals operate. There have been many structural changes in the Nigerian economy over the last decade; these include financial services, multinational firms, and global supply chains that have established organisational settings where decision-making is expected to be based on data. All these factors have created new needs in terms of managing and analysing financial data in organizations. This includes the financial inclusion project by the Central Bank of Nigeria (CBN,2012, p.1), and the establishment of the Treasury Single Account among public sector organizations (Odewole, 2016, p. 328-335).

At the same time, Nigeria's digital infrastructure presents distinctive challenges. Internet penetration remains uneven. Access to high quality connectivity, modern computing equipment, and premium software subscriptions is not uniformly available. In this respect, infrastructure becomes relevant in determining the analytics readiness problem, for instance, a lack of internet connection would impede the ability of professionals to engage in cloud-based analytics solutions, while organizations that do not offer premium software subscriptions will be forced to adopt freeware solutions. Lastly, professionals within organizations with a low level of digital transformation will have few chances to implement analytics skills in practical work situations.

The Nigerian professional accounting environment is largely defined by two major organizations, namely, the Institute of Chartered Accountants of Nigeria (ICAN) and the Association of Chartered Certified Accountants (ACCA), which feature significant Nigerian representation. The Institute of Chartered Accountants of Nigeria holds significant importance regarding the formulation of professional standards, examinations, and CPD. The degree to which ICAN and ACCA have incorporated practical data analytics competencies into their qualification programmes and CPD frameworks is therefore highly consequential for the analytics readiness of the Nigerian accounting workforce. The evidence from this study suggests that both bodies have significant scope to strengthen their analytics related provisions.

The post pandemic era hastened the digital transformation of companies in Nigeria. The COVID-19 pandemic compelled many firms operating in Nigeria to adopt telecommuting, digital accounting, and online accounting procedures at a much quicker rate than might have otherwise happened, thereby increasing demand for accountants who can work in digital accounting systems. As a result of this growing trend, there is a gap between the analytical skills required in many organizations in Nigeria and the type of education that most accountants received during their schooling. The study was carried out at the start of 2026, a period when this gap had considerable importance.

There is also another aspect to this issue. The level of competition faced by accounting professionals in Nigeria has become more pronounced with the advent of automation and artificial intelligence in the field of finance. ERP systems automate several transactional processing activities that had previously been done manually, and machine learning algorithms are now being employed in the auditing and risk management fields. Although this does not mean that there will be no requirement for accounting professionals, it does change the nature of the contributions that professionals make from being purely transactional to being primarily analytical, from report writing to translating raw data into meaningful information, and from compliance monitoring to performance recommendations.

2. Theoretical Framework and Literature Review

2.1 The Data-Driven Transformation of the Accounting Profession

The changes occurring in digital technologies along with the rising volume of available data led to significant transformations of the accounting field. Currently, organizations generate huge volumes of data, both financial and non-financial, using various enterprise systems, digital platforms, and other sources of operational data. Accounting, thus, does not limit to mere recording of financial transactions but also includes a significant part related to analysis and interpretation of this data (Bhimani and Willcocks, 2014, p. 475-477).

This phenomenon, according to Quattrone (2016, p. 119-120), represents reconfiguration of the practice of accounting triggered by digitalization where data is highly utilized by decision makers. This process is especially relevant for management accountants and financial controllers, as their profession traditionally consists not only of reporting activities but also analytical tasks. Asking them not only to report but also interpret and model the significance of certain financial data implies significant change in the professional mandate.

Bhimani and Willcocks (2014, p. 477-480) characterized this transition as moving from intuition and precedents to systematically guided by data and analytics accounting decisions based on thorough analysis of large, often complex datasets. However, the transition should be understood not merely in technical terms but also organizational and even cultural, as accounting professionals have to acquire new skills, learn to use novel tools, and work together with new stakeholders who would use accounting professionals' insights based on analysis of the financial and non-financial performance data.

In support of the idea that the transition to new skills occurred, Appelbaum et al. (2017, p. 30) provided empirical evidence showing how the integration of data analytics and enterprise systems has transformed the practice of managerial accounting and increasing demands to the ability of accountants to provide analytical insights in real time instead of relying on historical data only. Traditional competence model consisting of accounting expertise, professional ethics, and procedural accuracy is no longer enough on its own.

However, such a transition to advanced analytics intersects with other recent trends within the discipline of business administration, including rise of the performance management systems, integrated reporting frameworks, risk management, and ESG (Environmental Social Governance) issues. ESG has led to broadening the accounting and reporting scope, as accountants are now asked to work with non-financial data and its interpretation in terms of performance (Raghavan, 2022, p. 9).

However, a tension exists throughout the whole discussion the gap between emergence of the new tools of data analytics and readiness of accountants to utilize them properly. According to Sutton et al. (2016, p. 68-69), technologies advance faster than the skills of accountants, resulting in ever-growing disequilibrium between job demands and competencies of workers. This pattern was also noted by Granlund (2011, p. 2-5), institutional rhetoric on application of the tools of data analytics in accounting often outstrips their actual utilization as accountants continue to rely on more basic analytical tools. This mismatch is the reason for choosing this topic for research.

It is important to mention that the topic acquires special importance when discussed in regard to Nigeria a country where accounting industry experiences challenges peculiar to this particular context. Being among the largest African economies, Nigeria faces increasing needs for accounting professionals capable of performing analysis because of its rapid economic development, rise of foreign investments, and increased organizational complexity (Iyoha and Oyerinde, 2010, p. 364-365). At the same time, the local accounting environment is characterized by limited access to analytical tools, weak professional development programs, and underdeveloped education system that fails to incorporate analytical training systematically into education programs (Fijabi and Lasisi, 2023, p. 71-79).

2.2 Data Analytics in Accounting Practice

2.2.1 Defining Data Analytics in the Accounting Context

Data analytics entails an array of analytical tools and approaches, which are relevant for both financial and non-financial data. According to Appelbaum et al. (2017, p. 31-33), in the context of accounting practices, data analytics involves statistical, mathematical, and computational methods to derive valuable information from data, identify trends and anomalies in them and assist in evidence-based decision making. The approach includes three dimensions of analytics, descriptive (describing what happened, predictive (projecting what will happen) and prescriptive (suggesting what needs to be done). Application of these different dimensions within professional practice varies greatly depending on the specific job profile of a professional as well as his/her analytical skills.

As far as the tools relevant for data analytics in accounting are concerned, these vary from generic spreadsheet packages such as Microsoft Excel with pivot table, power query and data modelling features to advanced business intelligence tools such as Power BI, Tableau, Qlikview, statistical software like R and Python, enterprise ERP Analytics Modules, among others (Appelbaum et al., 2017, p. 33). Every dimension requires different skill sets and implies different analytical capacities for a professional (Appelbaum et al., 2017, p. 42).

2.2.2 Evidence from Empirical Studies

A growing body of empirical research has examined how data analytics is used in accounting practice. In the field of management accounting, Granlund (2011, p. 4) observed the expectation of an analysis of performance figures, creation of dashboards, and budgeting/forecasting, with the realization that practice frequently falls behind institutional expectations. With regard to auditing, Yoon et al. (2015, p. 432) show how analytics allows auditors to review all transactions in their population rather than sample selections, thereby altering the very essence of audit evidence and anomalies. From this, one can conclude that although the rationale for using analytics in accounting is clear, practice has been patchy.

Unevenness remains another common thread. Côrte-Real et al. (2017, p. 381-383) noted the role of organizational size, technology readiness, and management's commitment as key variables that influence the degree of usage of analytic tools within organizations. When it comes to individual differences, Richins et al. (2017, p. 74-75) pointed out the persistent mismatch between the increasing importance of analytic capabilities in the

definition of accountants' roles and their level of self-assurance in utilizing sophisticated analytic methods. This mismatch is especially evident in the lack of access to professional training which is typical of emerging markets such as Nigeria.

Specifically in relation to the professional environment of Nigeria, it has been found by Fijabi and Lasisi (2023, p.78-79) that lack of access to current software and training facilities substantially limits the use of data analytics within accounting practices. What seems quite clear from their research is that the environment of the professional sphere, in which such data analytics need to be used, presents its own set of challenges beyond individual shortcomings.

2.2.3 The Skills Gap in Data-Driven Accounting

One of the most significant findings from the empirical literature is that there exists a skills gap in regards to what the expectations are for the analytical skills of accountants compared to their current ability. Richins et al. (2017, p. 74-75) documented that while accounting professionals increasingly recognise the importance of analytics to their professional roles, they lack confidence in applying advanced analytical techniques in practice. Appelbaum et al. (2017, p. 39-42) identified this gap as a significant impediment to the effective integration of analytics into accounting functions, arguing that organisational conditions and individual capabilities must both be addressed for analytics to fulfil its strategic potential.

The skills gap is dynamic. With technology in analytics progressing at a faster rate, the disparity between the capabilities of the technology and those of the people who use it may become even greater if professional development lags behind. Sutton et al. (2016, p. 65,68) observed that the speed at which new analytical technologies are introduced creates a persistent challenge for professional preparation, since educational and training programmes typically adapt more slowly than the technology landscape evolves. This observation is particularly relevant in emerging economy contexts, where the educational and professional development infrastructure may be less responsive to technological change than in settings where such infrastructure is more robustly developed.

2.3 Professional Competence Theory

2.3.1 Foundations and Core Concepts

Professional Competence Theory represents the most fundamental theoretical perspective from which this research study explores the readiness of accounting professionals to engage in data driven tasks. It originated in the areas of occupational psychology and professional education and was extensively used throughout numerous professional domains such as medicine, law, engineering, and business. In accounting, the professional competence theory provides researchers with a conceptual model for examining not only the competencies of accounting professionals but also their perception of those competencies as well as their self-assessment of their own preparedness to handle the challenges associated with their professional role.

Mulder (2014, p. 111) describes professional competence as an integral combination of knowledge, skills, and the ability to use both in the effective manner in the context of specific work situations that may be characterized by high levels of complexity. There

are three key aspects that should be considered as noteworthy about his description. First of all, the author emphasizes the idea that competency is a process and, hence, it should not be viewed as the simple sum total of individual skills. Second, he draws the reader's attention to the fact that professional competence depends on context: it cannot be discussed as a universal notion but should always relate to specific professional activities. Finally, competence may vary from time to time due to changing demands for particular roles.

The key element in the professional competence model proposed by Mulder (2014, p. 114), is the notion of perceived readiness. According to the author, this term refers to self-assessment of individuals in terms of their ability to perform professional duties. While it relates to the actual competencies, perceived readiness is also characterized by a subjective element that makes this notion theoretically interesting as well as practically significant for professionals. For example, an accounting specialist may have a high level of proficiency in his or her work but may be unprepared to use the acquired knowledge and skills in new situations. At the same time, another professional may underestimate the complexity of the situation and overestimate his or her own abilities.

2.3.2 Applications to Accounting and Data Analytics

In the sphere of accounting, professional competence theory has seen its most extensive application in the examination of the technical skills of the accountants themselves, as well as their proficiency in ethical and professional judgement. As stated by Burns and Baldvinsdottir (2005, p. 749,751), drawing from theoretical models of competence, there was noted a shift in the skillset needed of management accountants, as the transition from mere financial reporting to true business partnership implied a change in the competence profile. In other words, this research, which had been performed via the lens of institutional theory, indicated that the new competencies demanded of accountants included analytical and strategic judgement skills along with their expertise in accounting.

Speaking about the relationship between data analytics skills and professional competence theory, it would be right to note that such a relationship is relatively new. The data analytics skills could be treated as one of the dimensions of the competence of accountants in modern times. It is supported by the fact that, according to the model of professional competence, the required skills vary with the evolution of the organisational and technological environment. In this case, the study draws on the above-mentioned concept of competence as it focuses on the factors that shape perceptions of readiness for analytics tasks in accounting firms.

2.3.3 Strengths, Limitations, and Critical Engagement

Professional competence theory provides several notable strengths in the current research design. First, the theory's focus on perceived readiness as an important aspect of professional competence allows for exploring the experiences of professionals in regard to their personal readiness for the performance of new tasks. Considering the fast-paced role transformation explored in this research, the gap between credentials and capabilities becomes more significant since a credential does not always correspond to one's actual competence in certain areas. Therefore, perceived readiness is particularly important in the study in question since it allows for evaluating whether people with various credentials see themselves as capable of performing advanced data analytics tasks

successfully. Another strength associated with professional competence theory lies in its consideration of the specifics of the particular work environment. Since the study focuses on the situation in Nigeria, which differs greatly from developed countries in terms of enabling conditions, this feature of the theory can be applied quite effectively in the process of explaining observed phenomena.

At the same time, there are several limitations of professional competence theory associated with the subject matter of the current research. First, this theory implies some difficulties in specifying the relevant competencies in rapidly changing professional environments (Mulder, 2014, p. 125). Indeed, in the context of the fast evolution of the analytics field, some skills needed to be perceived as necessary ones could be out of date by the moment they are identified in official assessment frameworks. Second, since professional competence theory is focused mainly on individual aspects, organisational and structural issues could be neglected, which could distort findings related to the level of professionals' competence development in the Nigerian setting. To address this problem, this study applies human capital theory and role theory alongside with professional competence theory.

2.4 Human Capital Theory

2.4.1 Foundations and Core Concepts

Human Capital Theory, founded upon seminal work done by Becker (1993), constitutes another theory that helps us gain additional perspectives on how accountants get equipped with skills that are critical in the analytics-driven era. Human Capital Theory postulates that knowledge and skills represent productive investments that people constantly make in themselves much like other physical resources and earn returns from in terms of professional productivity, employment and salary. Becker (1993, p. 33-40) distinguishes between inventory of human capital an asset accumulated through formal training and practice, and returns on this investment, which depend on how effectively those skills were used for productive purposes.

The differentiation between general and specific human capital is important in human capital theory. The difference between the two lies in the fact that general human capital represents skills that can be applied broadly, while specific human capital involves skills of use in one organisation only (Becker, 1993, p. 40). This differentiation is particularly relevant in terms of analytics skill building among accountants. Formal university education in accounting develops general human capital of foundational skills of accounting, financial reporting and quantitative analysis. On the other hand, analytical skills needed within a specific professional position for example, skills of using specific ERP systems of a certain employer, and producing certain dashboards requested by management teams mostly involve specific human capital that needs to be acquired through practice.

Following Nafukho et al. (2004, p. 546), human capital theory can also help highlight the fact that returns on investment in human capital greatly depend on how relevant those skills are for the current tasks. The fast-paced technological change implies that the human capital built up as a result of conventional accounting training might be rendered partially irrelevant regarding skills related to analytics. As such, continuous investments

are needed in analytics-oriented human capital, which is exactly what is provided for by findings of this study.

2.4.2 Human Capital Development Pathways

Human capital theory takes into account the fact that investing in skill development may be achieved through various means such as formal schooling, employer-provided training, personal learning, and experiential learning. The distinguishing features of each method are cost, availability, level and scope of human capital developed, and appropriateness for the existing job requirements. Becker (1993, p. 33-36) argued that training offered by the employers will create role-specific human capital, which increases the productivity of the individual within the organization itself, whereas the formal education of an individual creates more general human capital, which can be used anywhere.

Regarding the development of skills in the field of analytics for accountants, the literature highlights that the main channel through which skill development takes place in professional practice is informal, as opposed to the formal one. According to Richins et al. (2017, p. 63,74), analytics skills have become a crucial part of the human capital of accountants, although formal education programs have taken a considerable amount of time in adjusting to the inclusion of the practical skills of analytics in their curricula. The result of the delay in adjusting the formal education systems is that practitioners are left with two options either develop the skills by themselves, or remain without them.

It is also possible to use the theoretical framework to explain the variations in analytics readiness that exist within the professional population despite similar levels of human capital amongst the individuals. According to Nafukho et al. (2004, p. 546-549), the benefits of investing in human capital do not only rely on the amount of investment made but on the relevance of the investment relative to job-related demands and the environment that allows the use of skills gained from such investment. The above explanation aligns with the empirical results of this study which show that experience and educational qualifications are insignificant factors of analytics readiness.

2.4.3 Strengths, Limitations, and Critical Engagement

One of the strengths of human capital theory as applied to this topic is the understanding that there are many different ways to develop skills, and these methods may lead to varying degrees of effectiveness in terms of developing professionals' abilities. For that reason, it is very useful for determining the reasons behind the discrepancy in readiness among various professionals and for assessing how much of an investment in developing their skills would be required to narrow down the gap between what is expected from them and their actual abilities. Another benefit provided by human capital theory is its focus on the returns on investments rather than just the amount invested in skills development, which is especially useful given the prevalence of formal qualifications in the Nigerian job market that do not lead to analytics readiness.

There are a number of limitations associated with the human capital theory, and one of them involves the notion that skills development is predominantly seen as an individual process during which individuals decide independently whether to invest in their skills or not, based on the potential benefits (Nafukho et al., 2004, p. 546-547). In reality, the

ability to develop skills is affected by organizational culture, employers' decisions about investing in their employees, and other external factors that impact professionals. The limited access to training, computer programs, and professional development in Nigeria further complicates the picture. To address this limitation, it was decided to investigate the impact of organizational support on readiness and apply the role theory framework to this issue.

2.5 Role Theory and the Evolving Role of Accountants

2.5.1 Foundations and Core Concepts

The concept of role theories pertains to the definition, maintenance, and change of roles within both social and organisational environments. The tradition of researching roles from such a perspective is quite long-standing in both social psychology and organisational behaviour fields, where this topic has been studied since the work of Merton (1957), Katz and Kahn (1978), and Biddle (1986) among others. Within the context of the organisational environment, role theories refer to the role-related expectations of both external stakeholders, like clients and institutions, and internal actors such as colleagues and the organisation itself, alongside their understanding of such expectations.

A functionalist theory of roles has been presented by Biddle (1986, p. 70-74) as a description of how roles represent certain expectations set on people by institutional, organisational, and technological settings and how roles can evolve due to changes within such settings. This is a theory that applies directly to the current case of accounting professionals in digital times the institutional expectations of the profession are changing along with data-driven changes within organisations and their decision-making process.

There are two role theory-related concepts that prove especially relevant to the research topic of this study. The first one is the role conflict, defined by Biddle (1986, p. 82) as a situation when role incumbent encounters contradictions of the expectations that different stakeholders hold towards him or her. Role conflict can manifest in the case of an accounting professional, for example, when his/her employer or professional body sets expectations regarding the production of complex data analytical outputs while the person has not been prepared for that by educational and training institutions. The other concept is role ambiguity, defined as uncertainty about expectations associated with the role, which occurs in a situation where expectations are unclear to the role incumbent (Biddle, 1986, p. 83).

2.5.2 The Transition of the Accounting Role

This evolution of the role of accountants in organizations from financial reporters to analytical partners is well-documented in the extant literature. As some of the first scholars to investigate this issue, Granlund and Lukka (1998, p. 164-165) have shown how the process has unfolded, noting how accountants' role has shifted from mainly accounting and reporting to one involving increased participation in planning, controlling, and strategy decisions. Following up on this analysis, Burns and Baldvinsdottir (2005, p. 738) have gone on to apply institutionalism to the study of changes in management accountants' roles, specifically as they become integrated into cross-functional management teams thanks to both technological possibilities and organizational demands.

Burns and Scapens (2000, p. 13) have further argued that such shifts should not only be understood as path dependent and institutionally contingent, meaning that they will differ significantly from context to context.

One of the earlier empirical studies on the discrepancy between the formal role descriptions of management accountants and their actual behaviors on the job is Hopper's (1980, p. 409). According to his results, most managers' actions would not align with their job description because they had often assumed a 'servant' rather than an 'advisory' role. Such a gap between formal descriptions and enacted behaviors may foreshadow the dynamics identified in contemporary studies of analytics adoption, in which professionals may ostensibly be responsible for analytics work yet remain engaged in transactional and reportorial duties.

2.5.3 Strengths, Limitations, and Critical Engagement

The main advantage of role theory in relation to the present research is in its ability to link the experiences of an individual with regard to his or her perceived readiness within a broader structural analysis of the professional role transition process. Unlike approaches that attribute unpreparedness to some deficiencies on the side of the employee, role theory focuses the researcher's attention on institutional and organizational factors that contribute to the rapidity and adequacy of role adoption and performance. This is especially significant for this study as the pace of role transformation in Nigeria could have surpassed the pace of professional preparation due to some structural factors rather than employees' deficiencies.

Nevertheless, one can note one more drawback of role theory in that it works well for defining the structure of the phenomenon when there appears the gap between professional expectations and preparation but not for explaining the processes of gap closure. Role theory shows what the problem is; however, it cannot explain how role capabilities are developed. The missing part is compensated in this study via the integration of role and human capital theories into one analytical framework. The latter explains the mechanism of developing role-related skills in response to changing role requirements.

2.6 Previous Empirical Research on Data Analytics in Accounting

2.6.1 Education-Focused Research

The most studied element of data analytics in accounting involves issues in the field of education, exploring ways to incorporate analytic skills into accounting programs, the analytic skills that need to be acquired for the profession, and the barriers involved in implementing data analytics into the educational process.

According to Lawson et al. (2014, p. 296-298), accounting curriculum design needs to give more thought to the long-term career preparation of students by including skills in analysis and decision making instead of just focusing on knowledge related to entry level accounting work.

The major limitation in researching educational approaches to data analytics in accounting for the current study involves the emphasis on anticipated over actual

competence. Curricular approaches to studying data analytics involve the preparation of prospective accountants for the future not the ability of contemporary professionals to utilize analytics skills. Moreover, such literature focuses entirely on formal education and does not consider practitioner learning outside of formalized training. This limitation is partially addressed by the current study.

2.6.2 Practice-Focused Research

Another smaller but growing stream of academic research considers data analytics as part of professional practice and analyses the real capabilities of using analytics in the work of accounting professionals employed within their professional activities. Such practice-based studies are conducted mainly in North America and Europe through professional surveys, focus groups, and case studies on organisations.

One of the broadest practice-based studies was conducted by Appelbaum et al. (2017, p. 35) who explored the role of data analytics and enterprise systems in the work of managers involved in accounting. According to the study results, the presence of analytics in the managerial work is increasing while there were still variations among organisations and individual employees concerning the extent of the application of analytics. The authors concluded that there was a consistent discrepancy between the needs for analytical abilities in organisations and the skills of professionals in this area.

It was claimed by Sutton et al. (2016, p. 68-69) that a more systematic approach towards the examination of developing the skills in analytics is required in the future to develop the relevant theory. They emphasised the lack of an empirical evidence base on practitioner analytics skills despite the extensive amount of scholarly discussion on this topic. It was pointed out that the practice-based studies on the development of analytical skills in emerging economies are currently absent, although the process under consideration may be completely different in such environments compared to the ones in North America and Europe.

2.6.3 Research in Emerging Economy Contexts

The current literature related to the application of data analytics in accounting practices in the emerging economy has very little content. The majority of research conducted into the topic focused on practices carried out in North America, United Kingdom, Australia and Western Europe, where conditions of using analytics in terms of availability of appropriate software and quality of education vary greatly from conditions in developing countries like Nigeria.

The results obtained by Iyoha and Oyerinde (2010, p. 365-366) are relevant to give some general idea about Nigerian accountancy profession with regards to structural gaps in Nigerian accountancy profession infrastructure, as well as varying standards of accounting practice across different organizations. The authors stress out that the process of accounting should be analyzed in a specific institutional context.

In addition to that, Fijabi and Lasisi (2023, p. 72-74) identified specific barriers that hinder analytics adoption in accountancy practice in Nigeria. These studies contribute to obtaining the necessary information about accountancy profession peculiarities of Nigeria that will be used for contextualizing the current study.

However, the lack of research into accounting analytics readiness in the Nigerian setting is one of the crucial problems to be solved since, on the one hand, Nigeria is an important emerging economy while, on the other hand, it has a specific professional environment in which accountancy may develop differently.

2.6.4 Synthesis: The Research Gap and Positioning of This Study

The review of the relevant literature provides evidence of two concurrent gaps that have been identified and need to be filled through the current research study. Firstly, there is a greater inclination towards educational empirical studies compared to practice-oriented empirical studies relating to data analytics in accounting. Secondly, all practice-oriented empirical studies have a common flaw as they consider developed economies as the main subject of investigation, overlooking emerging economies such as Nigeria.

To fill these two concurrent gaps, the current study uses a combination of practice-oriented mixed methods empirical approach, which is related to the field of accounting and finance and the professionals who work in Nigeria. Through the theoretical foundations of role theory, human capital theory, and professional competence theory, along with the use of convergent mixed method design that incorporates both survey data from 125 respondents and interviews of six people, this study presents a unique perspective on analytics readiness.

2.7 Conceptual Framework

Based on the above discussion on the three theoretical frameworks discussed above, the conceptual framework for this study includes four constructs namely data analytics skills, perceived readiness, skill acquisition pathways and organisational support. The development of the framework was iterative in nature and based on the interactions with the theoretical literature. The following sections discuss each construct.

First, data analytics skills were conceptualised as per the professional competence theory framework where data analytics skills were seen as one of the contemporary accounting skills that have become more important in recent times (Mulder, 2014, p. 111). Data analytics skills were conceptualised as encompassing technical skills involved in using analytics tools and methods, interpretative skills required to analyse and interpret analytical outputs and the communication skills required to translate the findings for the consumption of non-specialist stakeholders. The exact composition of data analytics skills differs depending on the roles and situation, but the skills required in this particular study are the ones needed to address the most common analytics demands faced by accounting professionals in Nigeria, identified via surveys and interviews in this study.

Second, perceived readiness was conceptualised following the professional competence theory definition as self-assessment of one's readiness to meet the analytics demands of the role (Mulder, 2014, p. 114). Perceived readiness is the key dependent variable in this research because it involves what individuals are able to do and how they perceive their ability to achieve analytics demands in their professional roles. The distinction between perceived readiness and objective competence is an issue that should be explored further.

Skill acquisition is a form of human capital development where investment in education and trainings yields returns at both individual and organizational level (Becker, 1993, p.

15-16; Nafukho et al., 2004, p. 545-548). In this study, three different paths were considered, namely formal education (including university degree programmes and professional certification programmes); professional training (employer-provided training, continuous professional development programmes and short courses); and independent learning (self-learning, tutorials online, experimentation and mentoring). The three different pathways lead to analytics preparedness differently, with some pathways being more dominant than others among the Nigerian accounting professional population.

Finally, organisational support has been included in this conceptual framework as a moderating variable affecting pathways to skill development and influencing the relationship between pathway selection and perceived readiness. Based on the organisational dimension of human capital theory (Nafukho et al., 2004, p. 549-550) and role theory's view of role preparation (Biddle, 1986), it is acknowledged in this conceptual framework that analytics skill development is shaped by the organisational context. Software access, training opportunities, mentoring, and a culture of analytics learning are organisational factors that can contribute to higher levels of perceived readiness.

The theoretical contributions of role theory to this framework are mainly seen in the treatment of data analytics skill development as a response to changing role requirements and the identification of analytics role as a key structural requirement for perceived readiness. Role theory provides two concepts of role conflict and role ambiguity that were used to conceptualise situations where a professional faces the expectation of performing analytics tasks without being sufficiently prepared. These relationships are made explicit in the analysis section where empirical findings were interpreted using the three theoretical perspectives.

Figure 2.1 represents the conceptual framework of the study below. This model demonstrates the interaction between skill development process and organisational support systems and their impact on data analytics skills that influence readiness. The integration of the notion of role theory in terms of developing expectations in this conceptual framework is due to the understanding that analytics readiness should be regarded as a developmental rather than a static characteristic that emerges in response to changing career demands.

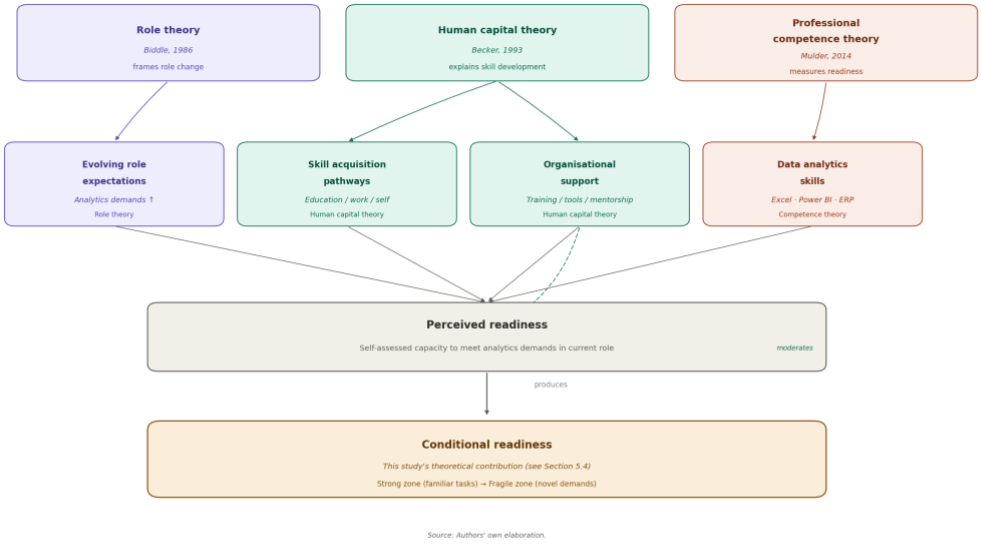


Figure 2.1. Conceptual framework of the study.

The conceptual framework forms the basis for the analysis undertaken in Chapter 5. In particular, in this chapter, the findings from the survey and interviews will be interpreted in light of each of the three theoretical perspectives in question successively. Moreover, it has also influenced the construction of survey and interview questions, which is discussed in Section 3.5. It is worth noting that this conceptual framework represents an integrative model, based on the complementary aspects of the three theoretical perspectives.

2.8 The Accounting Profession in Nigeria: Institutional and Historical Context

In order to be able to put the results obtained during this study in an appropriate academic context, it is first necessary to understand what makes the Nigerian accounting profession unique in terms of both its characteristics and the institutional environment in which the profession operates. Firstly, the origins of the Nigerian accounting profession lie in the period of British colonial administration, and even after the emergence of an independent state of Nigeria in 1960, the standards and norms of international professional communities played an important role in shaping Nigerian accounting practices. Furthermore, the institution established in Nigeria to manage these processes, the Institute of Chartered Accountants of Nigeria, founded in 1965, has made essential contributions to the development of accounting standards (Uche, 2002, p. 478-479)

The empirical description of the accounting environment in Nigeria was presented by Iyoha and Oyerinde (2010, p. 365-366). In particular, these researchers emphasized the shortcomings of the reporting environment structure, discrepancies in professional practices in different industrial sectors and the implementation issues associated with accounting standards in the public sector. Understanding accounting practice in its

institutional context turned out to be extremely important. Structural weaknesses mentioned by these authors included inadequate training infrastructure, insufficient access to information resources, as well as lack of incorporation of advanced analytical techniques in practice.

The shift from Nigerian Statements of Accounting Standards to IFRS entailed new challenges connected with the need for professional skills and financial data interpretation capabilities (Madawaki, 2012, p.156-157).

An update of the empirical picture of Nigerian accounting practices, which are now evolving due to digitalisation, was performed by Fijabi and Lasisi (2023, p. 71). These authors noted that lack of software tools and poor conditions for training accountants were important barriers to analytics implementation in Nigerian practice, and there were significant differences between levels of investments made in analytics infrastructure development in various sectors.

2.9 Emerging Economy Contexts and Analytics Adoption

Despite a growing body of literature on the topic, the interplay between digital transformation and accounting practice within emerging economies still requires substantial academic consideration. The Nigerian case represents an interesting example since the implementation of new technologies and innovations can be hindered by factors such as relatively low per capita incomes that limit organizations' ability to invest into training and development of their employees and individuals' access to professional development courses. Additionally, developing countries like Nigeria usually have less advanced digital infrastructures, which makes information systems less accessible and limits people's access to education and knowledge. However, despite these obstacles, there is no doubt that using analytics can help organizations enhance decision-making, improve risk management, and optimize operations.

The existing literature on accounting in sub-Saharan African states has identified several characteristics of the current environment that can have a considerable impact on the problem. First, it has been noted that the opportunities for pursuing further education and acquiring new skills are limited, which means that most practitioners engage in independent studies to obtain the knowledge needed to perform their job effectively. Second, it has been established that professional bodies play an essential role in creating the right working environment. For instance, ICAN and ACCA have a considerable influence on the development of the professional agenda for accountants practicing in Nigeria.

Finally, it should be noted that the emerging economy accounting literature contains several observations that can be useful for the current project. In particular, Iyoha and Oyerinde (2010, p. 362-366) have found evidence indicating that the link between educational background and practical competence can be significantly weaker in emerging economies compared to states where universities provide high-quality education and certifications guarantee specific abilities and skills. In this respect, it can be noted that the lack of correlation between the credential level and analytics readiness that was found by the present study is consistent with the observations.

2.10 The Role of Self-Directed Learning in Professional Development

Self-directed learning as the dominant mode of skill acquisition for Nigerian accounting professionals in analytics aligns with extensive scholarly literature concerning professional learning and development. According to Knowles (1975), self-directed learning denotes the process where individuals take the initiative in diagnosing their learning needs, setting objectives, accessing resources, and evaluating their progress. Self-directed learning has been shown to be an effective tool in skill acquisition when the necessary conditions motivational factors, self-awareness, and resources exist.

Accessibility and quality of learning resources determine whether self-directed learning is possible and effective. In the current context, the resources that accountants in Nigeria used in their efforts to develop analytics skills include YouTube tutorials, courses available from such websites as Udemy and CFI, and AI assistance tools like ChatGPT and Copilot. These are powerful and easily available sources of knowledge but they fail to provide a comprehensive perspective on developing analytics capabilities. For example, if a professional learns Excel pivot tables from a YouTube video, he or she will gain the ability to use those pivot tables successfully but will lack the knowledge about when pivot tables can be effectively utilized, what their limitations are, and how they relate to other types of analysis. This reactive character of self-learning in analytics is fully embodied in the conditional readiness pattern identified in the sample.

The relational components of professional learning should not be overlooked either. According to Lave and Wenger (1991, p. 98-100), learning occurs through participation in communities of practice that involve social interaction between novices and experienced professionals within the context of common professional activities. The social aspect of learning is apparent in the explanations of the interview participants: P2 emphasizes the value of mentorship alongside self-direction; P3 describes gaining knowledge by watching an analytically oriented supervisor at work. Furthermore, P2, P3, P4, and P6 mentioned organizational training programmes as part of their learning experiences. All of these represent forms of situated learning that surpass self-directed learning in their effectiveness because of the social component involved. Therefore, unequal access to such social learning options leads to uneven distribution of analytics readiness among Nigerian professionals.

Another key concept here is the self-efficacy theory formulated by Bandura (1986, p. 391). Self-efficacy can be understood as the belief in one's abilities to successfully perform certain actions. Self-efficacy plays an important role in people's ability to undertake self-directed learning activities as well as persist in these activities if obstacles arise. People who believe in their ability to succeed with analytics will tend to choose self-directed learning strategies and will be ready to put up with difficulties along the way; they will also seek mentoring help. The correlation between confidence in analytics and readiness observed during the chi square analysis (Cramer's V = 0.44) indicates a reciprocal relationship between confidence and readiness.

3. Methodology

3.1 Research Approach and Philosophical Positioning

The philosophical basis for this research is pragmatism. As stated by Creswell and Plano Clark (2018, p. 4), pragmatism entails the acceptance that in instances where there are multifaceted research problems, it would be more reasonable for the question to dictate the research method and not choose an epistemology from the beginning. This perspective is applicable in the present case since the research problem of assessing the extent to which accounting professionals are prepared for data-driven working involves seeking quantitative information regarding the predictors and prevalence of analytics readiness amongst a population of accounting professionals as well as gathering qualitative insight into the subjective nature of how such professionals perceive themselves as being ready for analytics in particular professional contexts.

An entirely positivist philosophy of science would allow one to statistically characterize the presence of the predictor and prevalence of analytics readiness in a representative population while not capturing those aspects of the construct that would relate to contextually contingent interpretive understandings of readiness a feature that is important both theoretically and empirically illustrated by the present research findings. On the other hand, an entirely interpretive approach would offer insight into individual experiences and perceptions, yet not provide an overview of the phenomenon across a population of professionals or allow one to identify any associations between readiness and its potential predictors.

Therefore, the philosophical justification behind pragmatism allows for a research approach whereby the research question guides and influences the choice of method and not vice versa (Bryman, 2016, p. 29). This position also justifies using a convergent mixed method design described by Creswell and Plano Clark (2018, p. 14) as a type where qualitative and quantitative data are simultaneously collected, analyzed using separate techniques, and interpreted together in order to achieve an enriched understanding of the phenomenon under study.

3.2 Research Design

A convergent mixed methods approach is adopted in this study (Creswell and Plano Clark, 2018, p. 67). In this form of research design, the quantitative and qualitative strands of research would be conducted simultaneously but separately in terms of data analysis and collection. However, there can be a convergence between the two strands during the process of interpreting the results through conducting a comparative analysis of the results from the two strands to see the extent of convergences, divergences and complementarities between the two strands. In the current study, a convergent mixed methods approach is more suitable since there are specific roles for each strand; that is, the quantitative strand aims to uncover patterns within a professional group while the qualitative strand provides a richer understanding of these patterns.

Figure 3.1 shows the overall research design, which indicates how the quantitative survey strand and the qualitative interview strand were conducted concurrently and integrated through the findings of the study.

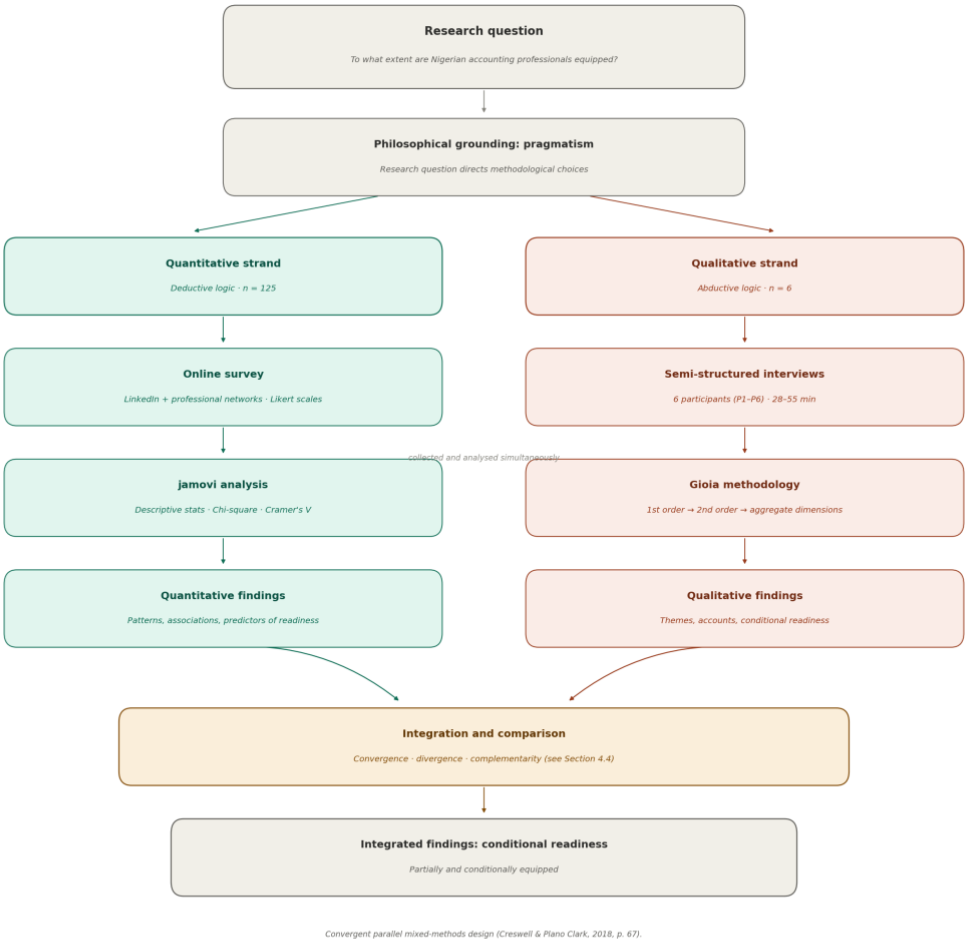


Figure 3.1. Research design overview.

For this study, a cross-sectional approach would be used, which entails collecting data once in order to obtain information on the current state of analytical skills readiness and associated factors for accounting professionals in Nigeria (Bryman, 2016, p. 62). The choice of a cross-sectional research design is based on the fact that the research being exploratory and descriptive in nature, it is sufficient to collect data only at the moment. The purpose is just to reveal the current state of readiness and its determinants without examining how these change with time. Not collecting longitudinal data is seen as a limitation of the study and indicated in Section 6.6.

Different research logics would be used in the two strands of this study. For the quantitative strand, a deductive logic is appropriate where constructs from professional competence theory, human capital theory and role theory would be operationalized to survey items and examined based on empirical data from a sample population of accounting professionals (Saunders et al., 2019, p. 124). The reason for using this approach in the current study is that the theoretical constructs used are clear and well-

defined and they include such constructs as perceived readiness, skill acquisition pathways, and organisational supports. On the other hand, an inductive with abductive elements of research logic is suitable for the qualitative strand of research according to the Gioia methodology (Gioia et al., 2013, p. 15).

3.3 Population and Sampling Strategy

3.3.1 Quantitative Strand: Survey Sample

The target population of the quantitative strand consists of accounting and finance professionals working in Nigeria. More specifically, the sample consists of various categories of accounting professionals such as financial accountants, management accountants, auditors, financial analysts, and finance managers. The target population is quite large and heterogeneous consisting of many different types of professionals working in many sectors of the economy, operating in companies of varying sizes, and located in different stages of their careers. Since a comprehensive and accessible sampling frame does not exist for the target population (there is no single register of all Nigerian accounting professionals), probability sampling is not an option (Saunders et al., 2019, p. 315).

Nonprobability purposive sampling was therefore employed. The criterion used to select participants is that they are currently working in accounting or finance in Nigeria or worked in one of these roles within the last two years. The criterion was met by selecting participants from the pool of individuals who received the survey sent via LinkedIn. In addition, personal communication with members of the Nigerian accounting community, such as accounting associations and alumni networks, was used. The survey was circulated in February and March 2026. The sample consists of 125 respondents, 92.8% of whom were based in Nigeria.

The sample cannot be considered a statistically representative sample of the entire target population of Nigerian accounting professionals. The use of LinkedIn for distributing the survey may have led to bias towards those professionals who are more active on this platform, as well as more accustomed to using analytics software. In other words, accounting professionals without LinkedIn accounts, individuals working for small organisations that lack technological resources, and those residing in geographically remote regions, may be underrepresented in the sample. These problems are further discussed in Sections 3.7 and 6.6. Despite the shortcomings, the sample is sufficient for meeting the goals of analysis and represents a professionally varied sample base for the research purposes.

3.3.2 Qualitative Strand: Interview Sample

In total, six accountants and financiers were selected for semi-structured interviews via purposive sampling. The inclusion criteria for participation included current employment within an accounting or financial capacity, having at least eighteen months of work experience, agreeing to talk about the use, readiness and skill development around analytics. Moreover, diversity in terms of professional role, industry and career stage were taken into consideration. The six individuals were chosen based on the heterogeneity principle recommended by Gioia methodology instead of statistically representing cross

section, various perspectives should be included through participant recruitment (Gioia et al., 2013, p. 16).

As a result of sampling, there have been selected P1 general accountant in manufacturing with almost three years of experience, P2 payroll analyst in banking and fintech with more than six years of experience, P3 internal control officer in the automobile industry with almost four years of experience, P4 general ledger accountant in manufacturing with almost five years of experience, P5 accountant in export and shipping company with almost two years of experience and P6 finance assistant in printing industry with less than two years of experience.

3.4 Data Collection Methods

3.4.1 Online Survey

Quantitative data will be gathered through a structured online survey using Microsoft Forms from February to March 2026. In general, the survey is expected to take 15 to 20 minutes and the pilot version of the questionnaire was completed by two accountants prior to final distribution of the survey. Two survey questions were identified to be ambiguous during the pilot testing process and revised versions of these questions were prepared prior to distributing the survey questionnaire to the sample group. Pilot testing has confirmed that the survey can be easily accessed using a desktop computer or smartphones, which is crucial for the targeted population since they may not be having same level of computer literacy.

In total, the survey consists of six main parts apart from the introduction section where the aim of the survey will be stated and voluntary participation as well as informed consent will be confirmed from the participants. Six main parts include: 1) Demographic characteristics and professional background; 2) Frequency of usage of different data analysis tools; 3) Skill level with various data analysis techniques; 4) Readiness and capability of developing required skills; 5) Skill acquisition strategies and 6) Challenges in developing and applying analytical skills. Questions were mainly developed in five point Likert scale forms, which are commonly used for measuring perception of attitudes in social research (Bryman, 2016, p. 166).

Survey questions were formulated according to the theoretical frameworks mentioned in Chapter 2. Questions regarding perceived readiness have been developed based on the theory of professional competence (Mulder, 2014), skill acquisition questions have been developed based on the human capital theory (Becker, 1993), while role responsibility questions have been developed based on the role theory (Biddle, 1986). Supervising team members approved the survey questionnaire and participant information sheet prior to conducting the survey.

3.4.2 Semi-Structured Interviews

Qualitative data were collected through six semi structured interviews conducted between 6 March 2026 and 4 April 2026. Interviews were conducted using Microsoft Teams as a virtual tool and lasted between twenty eight and fifty five minutes. Semi structured interviews were selected as the method of collecting qualitative data since this technique offers flexibility in probing into the participant's response without failing to provide the

necessary thematic framework to address all the research questions (Saunders et al., 2019, p. 437). Such flexibility becomes particularly pertinent when conducting interviews in order to interpret quantitative data.

An interview guide was developed based on the theoretical framework and major themes that emerged during the literature review. The guide consisted of four major thematic clusters: (1) changes in the professional role and analytic responsibility; (2) experiences in conducting analytics assignments in everyday professional practice; (3) processes of developing the required skills, including the role of formal education, on-the-job training, and self-study; and (4) organizational support and barriers to the development and implementation of analytics skills. Each cluster comprised an opening question followed by a series of probes for follow-up. The guide was reviewed and accepted by the supervisor before the interview stage.

All interviews were recorded with consent of the participant. Transcriptions of the interviews were produced and checked for accuracy in comparison with the audio files before proceeding to analyze the data. All participants were coded anonymously in line with ethical considerations outlined in Section 3.8 in all forms of output produced throughout the study. Identifying information is not contained in the research dataset at all.

3.5 Operationalisation of Key Constructs

The operationalisation of the five main theoretical constructs is provided in Table 3.1 below, with details of the theoretical foundations of each theoretical construct, the survey questions or interview topics used to measure each construct, and the scale used to measure each construct. This operationalisation allows each theoretical construct to be systematically converted into empirical variables, thereby making it possible to interpret the results in light of the theoretical framework developed in Section 2.7.

Table 3.1. Operationalisation of key constructs.

Construct	Theoretical basis	Survey items / measures	Scale
Data Analytics Skills	Professional competence theory (Mulder, 2014)	Self-rated proficiency: data cleaning, visualisation, interpretation, dashboard development, strategic data use	5-point Likert (1=low, 5=high)
Perceived Readiness	Professional competence theory (Mulder, 2014)	Perceived adequacy of preparation; confidence using data; perceived importance of analytics; perceived quality of education	5-point Likert (1=strongly disagree, 5=strongly agree)

Construct	Theoretical basis	Survey items / measures	Scale
Skill Acquisition Pathways	Human capital theory (Becker, 1993; Nafukho et al., 2004)	Primary pathway: formal education, structured professional training, self-directed/on-the-job, or none	Categorical (single selection)
Organisational Support	Human capital theory; role theory (Biddle, 1986)	Perceived sufficiency of organisational support for analytics skill development	5-point Likert (1=strongly disagree, 5=strongly agree)
Role Expectations	Role theory (Biddle, 1986; Burns and Baldvinsdottir, 2005)	Perceived changes in professional role toward analytical work; qualitative interview themes	Qualitative (semi-structured interview questions)

Source: Authors' operationalisation based on Mulder (2014), Becker (1993), Nafukho et al. (2004), Biddle (1986).

3.6 Data Analysis

3.6.1 Quantitative Data Analysis

Survey data were uploaded to jamovi (Version 2.6.45; The Jamovi Project, 2024) to run analyses. Preliminary data checks were performed prior to the main analysis, which entailed checking the quality and completeness of the data collected. There was minimal missing data across all items in the survey. All 125 completed surveys were included in the analysis (Field, 2018, pp. 46-49). Frequencies and percentages were calculated for each of the demographics and survey questions. The described descriptive statistics form the base of characterising the sample in Chapter 4.

In order to investigate potential associations between the variable measuring perceived readiness for analytics-related tasks and several explanatory variables, chi-square tests of independence were conducted. The perceived readiness item was selected as the main dependent variable as it captures the concept of overall analytics readiness that constitutes the focal point of the study. For the purpose of performing chi-square analyses, all Likert scale variables were grouped into three variables: Disagree, Neutral, and Agree. Grouping the Likert items increased the size of cells and reduced the percentage of expected counts below five. It is recommended by Field (2018, p. 729) as a strategy for overcoming the problem with the expected count assumption when conducting chi-square tests. In accordance with his advice, all the Likert variables were collapsed into three categories to ensure that no more than 20 % of expected counts fall below five.

Chi-square was chosen as the analysis strategy because both dependent and independent variables are categorical regardless of whether they are collapsed into smaller groups.

Although the sample size of 125 is small, chi-square can still be used due to the attention paid to the expected count assumption (Field, 2018, p. 729). Effect sizes were calculated for each statistically significant association between variables. After grouping the Likert-scale variables into three categories, several chi-square tables still contained more than 20 % of expected counts less than five, which means that the results of these tables should be interpreted with caution, especially in terms of effect sizes and p-values (Chapters 4 and 5).

It could be noted that the choice of chi-square over other techniques such as logistic regression or SEM is related to the mixed-methods study approach applied here. Since the aim of the quantitative strand of research is simply to find patterns and relationships between variables in general terms without constructing any predictive models, chi-square will suffice as an analytic tool (Field, 2018, pp. 713-730). In a convergent design, qualitative findings are supposed to complement quantitative results to provide an in-depth explanation of the topic studied.

3.6.2 Qualitative Data Analysis

Qualitative interview data were analysed using the Gioia methodology (Gioia et al., 2013). This approach was selected for two reasons. First, it is particularly well suited to research that seeks to generate new theoretical insights from participant experiences rather than to verify pre-existing hypotheses: the Gioia methodology's systematic movement from participants' own language to higher-order theoretical categories is designed to maintain proximity to the empirical data while enabling theoretical contribution. Second, it provides a transparent and reproducible analytical process that strengthens the credibility of the qualitative findings by making the process of interpretation visible and accountable.

The analysis proceeded in three stages. In the first stage, following careful reading and re-reading of all six transcripts, first-order codes were developed by staying close to participants' own words and capturing the key meanings expressed. This process involved line-by-line coding of each transcript, with codes assigned at the sentence or utterance level. The aim was to preserve the participants' own framings and interpretations rather than imposing external categories. Approximately 120 first-order codes were generated across the six transcripts.

In the second stage, first-order codes were examined for conceptual relationships and similarities and grouped into second-order themes that represented more abstract patterns in the data. This stage involved greater analytical abstraction, moving from the participants' language to the researchers' interpretive categories. Second-order themes were developed iteratively, with codes compared across participants to identify similarities and differences in their accounts of role change, analytics practice, readiness, learning pathways, educational preparation, and organisational context. Fourteen second-order themes were identified.

For the third phase, second order themes were aggregated into dimensions that represented the essential conceptual framework of the results and served as a link between the qualitative analysis and theory. Six aggregate dimensions emerged, namely the changing role expectations within a data-oriented profession; use of analytics in professional work; preparedness and competence; development of competencies;

discrepancy between formal education and job requirements; and organizational support and hindrances. The complete Gioia data structure, representing the transition from first order concepts to second order themes to aggregate dimensions, is illustrated in Table 4.7 in Chapter 4.

For further improvement in the rigor of the analysis, both members of the research team independently coded the data and then compared their coding. In case of any disagreements in the interpretations, the issue was resolved by going back to the original transcripts instead of finding a middle point between the two researchers. This approach strengthened the confirmability and credibility of the qualitative findings.

3.6.3 Integration of Quantitative and Qualitative Findings

The integration of both strands took place during the interpretation phase and proceeded according to the convergent research design paradigm outlined by Creswell and Plano Clark (2018, p. 67). In particular, the findings obtained from the two methodologies were juxtaposed against one another in a rigorous process of comparison during which each of the major quantitative results was subjected to analysis, and the question was raised whether the qualitative findings confirmed, modified, extended, or contradicted it. Conversely, the same procedure was applied to all major qualitative themes: does this theme appear in the data set collected using the quantitative methodology, and, if yes, how?

The convergence in several aspects was discovered due to this methodological approach, including the use of informal channels for knowledge acquisition, the connection between organizational support and readiness, the lack of sufficiency of formal education in coping with analytics issues, and the effect of familiarity with tools on the confidence level. Moreover, an interesting divergence emerged between the moderately positive aggregated readiness scores obtained via the questionnaire survey and the more contingent and nuanced self-assessments provided by interviewees. This divergence is analyzed in detail in Section 4.4, serving as an empirical basis for the idea of conditional readiness introduced in Section 5.4.

3.7 Research Quality and Trustworthiness

3.7.1 Quantitative Quality Criteria

As far as reliability is concerned, with respect to the quantitative part, the standardized questionnaire with items in Likert scales based on established theories made sure that the reliability of the constructs being examined was assured. The internal reliability of the multi-item perceived readiness construct was tested at the pilot test phase. The construct validity is achieved through basing all survey questions on established theories and conducting a pilot test using two members of the target group before distributing the survey. This makes it possible to modify confusing survey items. The external validity, which is the generalizability of results to the larger population of accounting professionals in Nigeria, will be constrained due to the non-probability nature of sampling and possible selection biases caused by the mode of distribution through LinkedIn.

3.7.2 Qualitative Quality Criteria

Trustworthiness is examined here with regard to the four criteria of trustworthiness set forth by Lincoln and Guba (1985), further explained in terms of qualitative business research by Bell et al. (2019, pp. 363–365).

Credibility, i.e., plausibility and consistency of the story about social reality (Bell et al., 2019, p. 363), is ensured through the systematic three-stage coding technique used in the Gioia methodology and through the independent coding exercise undertaken above. Quotes made by direct interview participants will be used in Chapter 4 and help in assessing the credibility of the study since they will provide the reader with an opportunity to judge how credible the interpretations provided are.

Transferability is the extent to which the research findings can be generalized to other settings (Bell et al., 2019, p. 365). Since this study is based on a small and purposively selected sample of respondents, and since they belong to some particular sectors and are at different points in their career development, transferability can be considered fairly low. Thus, the findings should be seen as providing some insight but not as being representative of all Nigerian accounting professionals. However, detailed information about the characteristics of the sample and professional setting will be provided for those interested in transferring them to their own settings.

Dependability is "the ability to rely on the research process to produce consistent outcomes" (Bell et al., 2019, p. 364). It is achieved in this study through detailing the entire research process. For example, interview guide, method for obtaining consent from participants, coding process, and analysis of data are all recorded meticulously and systematically.

Confirmability is the extent to which research results depend on participant reports rather than researcher biases (Bell et al., 2019, p. 365). Confirmability is achieved in this study by clearly separating the language of participants and that of researchers in the process of analyzing interview results and interpreting them. The double-coding technique described in Section 3.6.2 will further improve confirmability since it will minimize idiosyncrasies associated with individual interpretation of data.

3.7.3 Mixed-Methods Quality

In order for the research that employs the mixed methodology approach to produce high-quality results, there should be consideration not only concerning the integration of the strands but also their quality (Plano Clark and Ivankova, 2015, p. 5). In relation to the current research, the integrity of the integration will be guaranteed by the process of comparing results typical of the convergent design; in particular, the findings produced through the use of the two research approaches will undergo cross-comparison and analysis.

3.7.4 Addressing Self-Report Bias

This research does not assess objective skill but instead uses self-reports and perceived readiness, which raises concerns about participants' systematic tendency to overestimate their abilities because of social desirability effects or, conversely, underestimate

themselves excessively due to high levels of self-criticism. To some extent, this drawback has been alleviated by conducting semi-structured interviews, which offered participants a chance for more sophisticated self-evaluation.

In this case, the difference in perception between self-reported higher levels of readiness from surveys and more cautious perceptions based on qualitative data represents one of the reasons why an aggregated self-report measure can produce a biased result. In addition, the distribution via LinkedIn could be seen as introducing selection bias among those people who are actively engaged in analytics and use online tools more often than the rest of the population.

3.8 Ethical Considerations

This study has fully met the requirements regarding ethics in the research process, which is stipulated in accordance with the requirements for the School of Business, Economics, and Statistics at Umeå University as well as by Saunders et al. (2023, p. 253) and Bell et al. (2019, p. 112). These requirements are aimed at the social sciences research and contain many important aspects, such as informed consent, participant privacy and confidentiality, transparency, integrity, and data protection rules.

The informed consent procedure was conducted by means of handing out the consent form together with the informational letter containing the information about the purposes of the study, the voluntary nature of participation, the freedom to refuse participation at any time, and the measures to preserve confidentiality before starting the interviews. Also, the interviewees orally agreed to participate during the introduction to the interviews that were then conducted. Before data collection, the supervising committee reviewed and accepted the consent form and the information letter.

Participants' privacy and confidentiality was secured by allocating the anonymous codes (P1-P6) to all participants before analyzing the results of the study. It should also be mentioned that only anonymous survey results without any identification details were obtained. The interviews were recorded and then stored on a safe server in university systems. Interviewees were also informed that their recordings and anonymized quotes would be used in the published dissertation.

To make sure that no potential harm would occur to the participants, they were allowed to decide on when and how to conduct the interview as well as receive the information about the topic of discussion and be assured that there were no correct or incorrect answers. During the interview, questions requiring any commercially sensitive or personally sensitive information were not asked, and all the information that was collected from the interviewees was related only to their professional experience.

This dissertation does not have any sponsors who could potentially affect the design of the research, its conduct, and reporting. Thus, there were no conflicts of interests. The research was carried out objectively without any biased attitudes or preferences caused by the researcher's personal value system and theoretical inclinations.

Concerning the application of AI software, any analytical, argumentative, and theoretical approaches used in this dissertation were determined by our own academic reasoning. Any use of AI tools was limited to writing support tasks and was conducted within the

boundaries set by Umeå University's code of academic integrity as described in the Canvas guidelines for this course..

3.9 Literature Search Strategy

The literature review that forms the basis of the research study has been carried out through a systematic search process of academic journals available in the databases of Umeå University Library, namely Business Source Complete, Scopus, and Web of Science. The keywords used for the search are "data analytics AND accounting," "accounting professionals AND digital transformation," "professional competence AND accounting," "human capital theory AND accounting skills," "role theory AND accounting," "data analytics AND emerging economies," and "accounting practice AND Nigeria." In addition, reference list searching has been used to complement the article search.

The research process took place in phases between January and March 2026. While the search process yielded a large number of documents, an extensive screening process on the titles and abstracts of the retrieved articles was carried out. Subsequently, a full-text analysis of those articles that had passed the initial screening phase focused mainly on empirical studies carried out within professional practice environments, theoretical literature related to the three adopted frameworks, and literature dealing with the accounting profession in Nigeria or emerging economies. The emphasis is mainly on scientific research articles published between 2000 and 2026, although some foundational texts such as Becker (1993), Biddle (1986), Hopper (1980), and Mulder (2014) were considered because of their theoretical significance in the relevant scholarly discussions.

In conducting the literature search process, there has been a deliberate attempt to give preference to empirical research articles compared to course books, in accordance with the instructions in the Umeå University Thesis Manual that higher-level theses must be based on scientific research articles and specialist academic literature. Course books such as Bryman (2016), Saunders et al. (2019, 2023), Creswell and Plano Clark (2018), and Field (2018) have only been used for methodological justification. From the findings of the literature search, the research gaps outlined in Sections 2.6.3 and 2.6.4 have emerged. These include over-research on educational issues to the neglect of practice issues and the absence of empirical studies concerning data analytics readiness in Nigeria.

3.10 Reflections on the Research Process

Conducting a convergent mixed methods study raises a number of practical and epistemological issues that merit reflection, both for the sake of transparency and recognizing the limitations inherent in the chosen path. There are three such reflections that stand out as especially relevant to the quality of this study.

The first of these pertains to the decision to distribute the questionnaire through LinkedIn. The lack of a complete sampling frame for Nigerian accounting professionals made it impossible to use probabilistic sampling, while LinkedIn was chosen as the channel with which to reach the largest possible number of Nigerian accountants who would be able to engage in the survey. While this reflects certain practical necessities, it does create a limitation that is recognized consistently throughout the thesis. The LinkedIn sample probably tends to include more digitally savvy individuals who are better connected to

international professional networks, and who may be more likely to adopt analytical thinking than the average Nigerian accountant. Future studies should strive to balance the LinkedIn sampling method with recruiting Nigerian accountants directly from the ICAN and ACCA professional networks, sampling firms and industries/regions underrepresented on LinkedIn.

The second pertains to the choice of methodology for qualitative data collection. While the decision was made to rely on the Gioia methodology, which was found suitable for creating theories from participant descriptions rather than confirming or disproving any hypotheses about their behavior. This means that there is no attempt to confirm the validity of the hypotheses set forth in the literature review and theoretical framework the purpose of this study is not to test any theories, but rather gain an understanding of the phenomenon of analytics readiness in the context of Nigeria. This is especially well reflected in the Gioia methodology's focus on using participants' language as much as possible in the first order coding stage, and the transparency of the interpretative process through the presentation of structured data in Table 4.7.

Thirdly, the combination of the two strands of the research at the stage of analysis entailed making some judgement concerning dealing with the inconsistencies between them. The biggest inconsistency that was noted between the fairly positive aggregated readiness scores obtained from the survey and the much more conditional and limited readiness evaluations made by the interviewees was dealt with by prioritizing the qualitative strand because it appeared to offer better and richer evidence regarding how readiness is perceived and experienced in certain contexts. This approach is theoretically supported by the idea of conditional readiness described in Section 5.4, although it must be admitted that another researcher could interpret the said discrepancy quite differently, for instance, attributing it to the social desirability factor.

4. Empirical Findings

The findings from this study are discussed in three parts in this chapter. First, the quantitative results from the online survey conducted with 125 Nigerian accountants and finance practitioners are presented, including full descriptions of the respondents' characteristics, descriptive statistics of tool use, self-rated competencies, perceived preparedness, skill acquisition sources, and the outcomes of the chi-square tests of independence to assess the relationship between preparedness perception and a variety of independent variables.

Second, the qualitative findings from six interviews conducted with participants using the semi-structured interview guide and analysed according to the Gioia method are reported, with the complete Gioia data structure provided in Table 4.7. Third, a synthesis of the two sets of results is undertaken, highlighting areas of convergence and divergence in the two bodies of evidence.

In this chapter, the findings do more than merely present results; they interpret the results in terms of what they mean in the context of the study's research problem. The quantitative results are placed in a wider context by the qualitative narratives, while quotes from the interviews help to explain and elaborate on the patterns identified in the quantitative results. In the course of the analysis, all participants are designated with an anonymous code (P1 to P6), following the principles outlined in Section 3.8.

4.1 Quantitative Findings

4.1.1 Respondent Profile

Table 4.1. Respondent demographic characteristics.

Variable	Category	Frequency (N=125)	Percentage (%)
Country	Nigeria	116	92.8
	Outside Nigeria	9	7.2
Role	Financial Accountant	42	33.6
	Other Roles	43	34.4
	Financial Analyst	15	12.0
	Auditor	14	11.2
	Senior Finance Roles	11	8.8
Experience	0-2 years	25	20.0
	3-5 years	55	44.0
	6-10 years	31	24.8
	11+ years	14	11.2

Variable	Category	Frequency (N=125)	Percentage (%)
Industry	Financial Services	44	35.2
	Manufacturing	18	14.4
	Public Sector/Education	14	11.2
	Consulting	14	11.2
	Other	35	28.0
Qualification	Bachelor's / HND	70	56.0
	Master's / in view	33	26.4
	Professional certification	17	13.6
	Other	5	4.0

Source: Authors' survey data (2026).

The total number of surveys used for this research amounted to 125. Out of them, all 125 were found to be usable. It was found that the overwhelming majority of respondents came from Nigeria (92.8%) although 7.2% of the surveyed were working outside Nigeria. Thus, these data should be considered complementary to the analysis of the data gathered in Nigeria. In respect to the position respondents were holding in organizations where they worked, 33.6% of them were financial accountants while the remaining respondents held other positions (34.4%), financial analyst positions (12.0%), auditor positions (11.2%), and senior finance positions (8.8%).

Concerning the experience in years, 44.0% of surveyed had 3-5 years of experience, and 20.0% had 2 years of experience or even less. 11.2% of respondents had 11 years of experience or more. These data show that a significant number of the surveyed had a rather low experience level. Given this factor, it is necessary to take into consideration this aspect when analyzing the findings about the readiness of the surveyed. Since the majority of the sample consisted of relatively inexperienced people, their analytical abilities were under development, and educational and training resources mattered.

Industrial distribution showed the highest share in the financial services sector (35.2%), manufacturing (14.4%), public sector/education (11.2%), and consulting (11.2%). Highest qualifications comprised a Bachelor’s degree or HND (56.0%), Master’s degree (26.4%), and professional certification (13.6%). The obtained information implies that the sample was educated but did not include many respondents holding postgraduate analytics-oriented degrees.

4.1.2 Data Analytics Tool Usage

Table 4.2. Frequency of data analytics tool usage.

Tool	Low (%)	Moderate (%)	High (%)
Advanced Excel	28.8	20.0	51.2
Power BI / Tableau	47.2	30.4	22.4

Source: Authors' survey data (2026).

From the tool usage results, an analytics tools hierarchy can be observed, which is both significant and theoretically grounded. Advanced Excel usage in terms of functions like pivot tables, Power Query, VLOOKUP, data validation was recorded highly frequently in 51.2 % of cases and moderately often in another 20.0 %. Hence, advanced Excel can be considered as the dominant analytics tool used by Nigerians, as far as accountancy practice is concerned.

Two factors make this result important. First, this high percentage proves that analytics has penetrated into the everyday accountancy work of a large part of our sample. Analysis of the data in question through pivot tables and creation of management reports from that data involve real analytics, not just routine operations. Second, this result implies certain limitations in terms of the tools' capabilities that we can observe here. The main analytics task accomplished using Excel appears to be descriptive and operational: data summarization, reconciliation, tracking of trends, and creation of management reports based on that.

Power BI and Tableau usage in this context appears to be quite low, as only 22.4 % of our sample used these analytics tools highly frequently, while 47.2 % used them infrequently. These analytics tools allow creation of highly engaging business intelligence outputs. As a matter of fact, usage of these tools shows that part of the sample engages in fairly basic analytics and not really the analytics that is the state of the art today.

4.1.3 Self-Rated Analytics Skill Levels

Table 4.3. Self-assessed analytics skill levels.

Skill Area	Low (%)	Moderate (%)	High (%)
Dashboard Development	38.4	40.0	21.6
Data for Decision-Making	23.2	34.4	42.4

Source: Authors' survey data (2026).

The differences in self-assessed skills show significant differences between the two types of skills analyzed. As mentioned earlier, the domain in which individuals felt the least competent was the construction of dashboard tools, where 38.4% reported feeling incompetent, 21.6% very competent, and 40.0% somewhat competent. In line with this, the use of tools showed similar results because the creation of a good dashboard tool needs knowledge in software like PowerBI or Tableau, which is rarely used by most participants in our study.

Strategic decision-making using data analysis was the strongest competency among those assessed, as most participants reported feeling high competency in this skill set, with 42.4% feeling highly competent and just 23.2% feeling incompetent. It is essential to note that the distinction between these two skill sets is theoretically crucial because it implies that analytics readiness is not a one-dimensional construct. Therefore, a professional can feel adequately skilled to use data to develop concepts, identify trends, interpret variances, make inferences about financial statements while remaining incompetent at executing technical tasks to create complex outputs.

This distinction leads to the notion of conditional readiness, which is discussed in detail in Chapter 5. In other words, this means that when analyzing surveys on readiness levels, it is crucial to understand that having most respondents feel adequately ready does not imply they feel ready to undertake all analytics activities, but rather that they feel ready for the particular tasks in their job profiles.

4.1.4 Perceived Readiness and Related Capability Statements

Table 4.4. Perceived readiness and related capability statements.

Statement	Disagree (%)	Neutral (%)	Agree (%)
I feel adequately prepared	12.8	25.6	61.6
I am confident using data	8.0	17.6	74.4
Analytics skills are essential	7.2	19.2	73.6
Education prepared me well	26.4	24.0	49.6
Organisation provides support	18.4	24.0	57.6

Source: Authors' survey data (2026).

Results regarding perceived readiness paint a relatively positive image. More than two-thirds of the participants (61.6%) reported feeling ready to handle data analytics tasks in their current jobs. The participants were more confident that they could use data to help the management make decisions, as 74.4% of them agreed on this statement. Furthermore, a significant percentage of the participants (73.6%) reported that knowledge of data analytics is essential to do their jobs effectively.

At the same time, three features of the results should be taken into consideration as qualifications to the generally positive picture obtained. First, there is a substantial difference in the proportions of respondents providing positive answers for all five statements. Two items about confidence of utilising data (74.4%) and the necessity of having analytics skills (73.6%) had considerably higher proportions of positive answers than the third one (61.6%) regarding perceived readiness. In other words, more respondents are confident about having general knowledge of working with data than being ready in terms of specific analytics. That may seem to be an intuitive conclusion; however, this differentiation appears to be quite subtle.

Second, the respondents' perception of their education preparedness revealed comparatively low results: 49.6% of respondents felt prepared due to their education while 26.4% disagreed with the given statement, and 24% remained neutral about it.

These numbers are associated with the highest proportion of respondents giving contradictory answers to the whole set of questions discussed in Table 4. At the same time, they show that education is not viewed as a major factor in developing readiness for analytics. Thereby, these results imply some specific considerations related to the development of accounting programs.

Third, organisational support is evaluated positively by a considerable proportion of respondents (57.6%), but it is the item with the highest proportion of disagreement. As many as 18.4% of participants did not agree that their organisation provided sufficient support in the development of skills needed for analytics, whereas another 24% remained neutral. In other words, there is almost half of the sample (42.4%) whose perceptions of the issue may be described as negative.

4.1.5 Skill Acquisition Pathways

Table 4.5. Primary skill acquisition pathways.

Pathway	Frequency	Percentage (%)
Self-learning / On-the-job	77	61.6
Structured professional training	32	25.6
No analytics skills developed	16	12.8

Source: Authors' survey data (2026).

The most surprising results are those related to the acquisition of skills in the context of the research. Over 60% of survey participants, precisely 61.6%, learned their analytics skills on their own and on the job, while 25.6% had learned the skills professionally via training sessions, whereas 12.8% said they did not acquire any analytics skills.

There are many substantive implications of the predominance of self-learning. It is great that professionals learn to be more analytical using online resources, such as tutorials in YouTube and other social media sites. Nonetheless, it is obvious that there is inconsistency in the scope and extent of self-learning. Professionals learn how to do things well, but these tasks happen to be mostly repetitive since they repeat them all the time. In turn, professionals lack sufficient understanding of what needs to be done in order to analyze something new. This kind of behavior fits the conditional readiness profile discussed above. Since professionals know they can do tasks they know very well, as they practice these tasks very often, they do not know whether they will be able to handle new tasks.

Finally, 12.8% of survey participants said they did not develop analytics skills. They are professionals who work in accounting and finance, and they operate in an environment in which the requirements for analytics are growing, and yet they did not even start to develop this competence. The reasons behind this phenomenon need to be explored further.

4.1.6 Challenges in Developing and Applying Analytics Skills

The survey also gathered data on the barriers for analytics readiness skill development and utilization. The barriers mentioned by respondents include a similar structure: lack of

structured training programs was considered the biggest barrier, followed by lack of access to sophisticated software packages, time pressure caused by workload and loss of previously gained skills due to infrequent use. The challenge pattern indicates the presence of personal motivation in the analytics readiness skill development and usage process; however, the necessary structural factors needed to convert this motivation into skills are lacking or underdeveloped.

There are two elements of the pattern of challenges worth mentioning in particular. The first point is the impact of infrequent use on the erosion of skills: professionals that have already mastered some analytics readiness skills will eventually lose these abilities because their jobs do not allow them to utilize this knowledge continuously. Therefore, there is an environment that makes it difficult to gain and maintain analytics readiness skills, especially if the person lacks organizational support. The other element is the emphasis placed on the problem of limited software availability as the biggest barrier to analytics readiness: this shows how the infrastructure inequality can affect analytics readiness, as individuals in firms equipped with analytics software have different skill sets than those in firms without such equipment.

4.1.7 Chi-Square Association Tests

Table 4.6. Summary of chi-square test results.

Variables Tested	chi-sq.	df	p-value	Result
Education x Readiness	58.65	4	<0.001	Significant*
Confidence x Readiness	47.83	4	<0.001	Significant*
Organisational Support x Readiness	42.64	4	<0.001	Significant*
Dashboard Skill x Readiness	36.27	4	<0.001	Significant*
Decision Skill x Readiness	34.27	4	<0.001	Significant*
Power BI Use x Readiness	28.75	4	<0.001	Significant*
Advanced Excel Use x Readiness	25.83	4	<0.001	Significant*
Learning Pathway x Readiness	22.96	4	<0.001	Significant*
Experience x Readiness	4.86	6	0.562	Not Significant
Role x Readiness	10.07	8	0.260	Not Significant
Qualification x Readiness	3.95	6	0.684	Not Significant

Source: Authors' survey data (2026). Note: Results marked with * should be interpreted with caution, as more than 20 % of cells in several association tables had expected counts below 5, even after variable grouping (Field, 2018, p. 729). This constraint does not affect the direction of the associations identified but limits confidence in precise significance thresholds.

The chi-square tests of independence showed that there are several factors that correlate with readiness for analytics-related tasks. The analysis demonstrated that the best predictors of readiness include capability factors and the context, while biographical factors do not show much correlation with readiness.

The best predictor of readiness among all tested factors was the perceived level of preparation through education, chi-square(4, N = 125) = 58.65, $p < .001$, Cramer's V = 0.48, with a moderately high correlation. It means that respondents who felt that they had enough education about analytics were more ready to perform analytics-related tasks. Nonetheless, it is important to note that the relationship cannot be understood as a simple confirmation that formal education creates analytics readiness. As the qualitative results indicate in Section 4.3, formal education is considered theoretically adequate but insufficiently practical. Thus, the more correct interpretation is that professionals who considered their educational background as adequate were in the best position to benefit from other learning opportunities provided by the organisational and personal contexts.

The next best factor that correlates with readiness is confidence in applying data to support decision-making, chi-square(4, N = 125) = 47.83, $p < .001$, Cramer's V = 0.44. This relationship confirms the importance of self-efficacy in relation to readiness, which is recognised by the professional competence theory as an aspect of perceived readiness. Thus, professionals who are confident that they have enough skills to make analytics-based decisions will likely feel more ready to perform tasks involving data-driven analyses.

Next, the organisational context shows a moderate correlation with readiness, chi-square(4, N = 125) = 42.64, $p < .001$, Cramer's V = 0.41. It means that the environment where professionals operate has a direct effect on how ready they perceive themselves to engage in analytics-related tasks.

Skill factors demonstrate some connection to readiness, although at a lower significance level. First, there is a connection between readiness and the skill to build dashboards, chi-square(4, N = 125) = 36.27, $p < .001$, Cramer's V = 0.38, and using data strategically, chi-square(4, N = 125) = 34.27, $p < .001$, Cramer's V = 0.37. Next, tool usage also demonstrates a connection with readiness. Specifically, Power BI and Tableau usage correlates with readiness, chi-square(4, N = 125) = 28.75, $p < .001$, Cramer's V = 0.34, and so does advanced Excel usage, chi-square(4, N = 125) = 25.83, $p < .001$, Cramer's V = 0.32. Finally, the way professionals learn and develop skills correlates with readiness, chi-square(4, N = 125) = 22.96, $p < .001$, Cramer's V = 0.30.

Interestingly, years of experience did not show a correlation with readiness, chi-square(6, N = 125) = 4.86, $p = .562$. Neither did current job title, chi-square(8, N = 125) = 10.07, $p = .260$, and highest qualification level, chi-square(6, N = 125) = 3.95, $p = .684$.

4.1.8 Summary of Quantitative Findings

The insights from the analysis reveal a clear image of the analytics usage by the Nigerian accounting profession to be one where analytics is well integrated into daily operations through the adoption of advanced Excel but with capability development being more pronounced at the basic than advanced level. Readiness for analytics is moderately high on average although individual differences exist in readiness and the factors that

determine it to be more linked to capability development and the facilitating work context than to career level or certification status. Individual and job-based learning play a leading role in capability development while structure-related obstacles are responsible for inhibiting analytics readiness among many professionals.

4.2 Qualitative Findings

Qualitative interviews were conducted with six accountants and finance professionals, and these interviews were conducted between 6 March 2026 to 4 April 2026. Different backgrounds were held by respondents in terms of professional, industry and organizational level perspectives, which have been discussed in detail in Section 3.3. As per the principles of ethical conduct mentioned in Section 3.8, all the participants will be coded here in this thesis (P1-P6).

For analyzing qualitative data of the interviews, Gioia's approach was employed (Gioia et al., 2013). The data analysis process included three main stages: coding first order concepts based on participants' own language; grouping of first order concepts into second order themes; and finally consolidating second order themes into aggregate dimensions. Figure 4.1 below shows the process model of data analytics readiness based on the Gioia approach, whereas Table 4.7 shows the Gioia data structure in its entirety. Six aggregate dimensions came out of the analysis, including: evolving roles expectations in data driven profession; application of analytics in professional work; readiness/capability gap perception; skill development strategies; misalignment in education and workplace requirements; and organizational enabling factors and constraints.

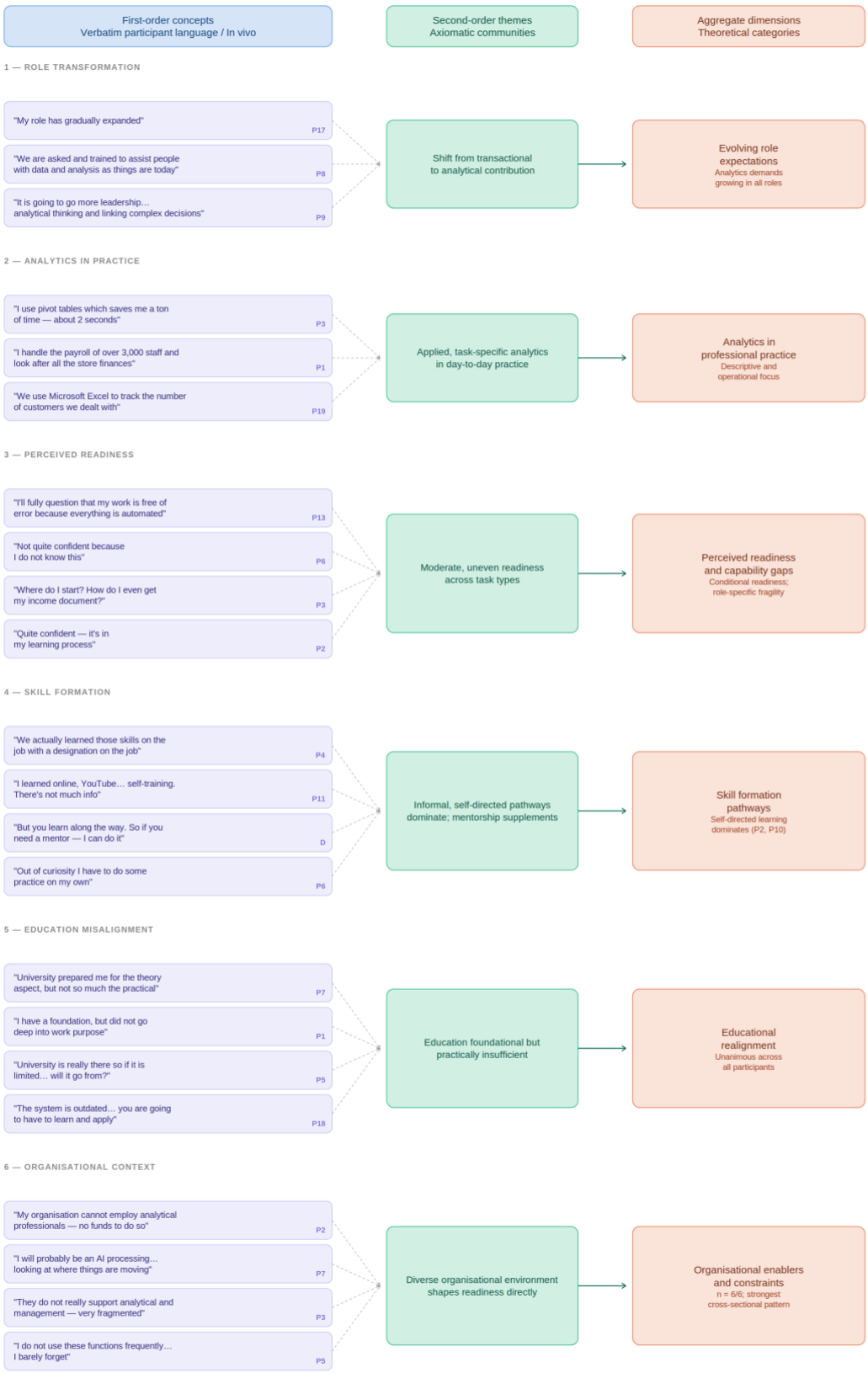


Figure 4.1. Gioia-based process model of data analytics readiness.

Table 4.7. Gioia data structure: first-order concepts, second-order themes, and aggregate dimensions.

First-order concepts	Second-order themes	Aggregate dimensions
Role has gradually expanded; no longer just following process; work with management; accounting will go beyond bookkeeping toward analytical thinking	Expansion of accounting roles into analytical and advisory work	Evolving role expectations in a data-driven profession
Data analytics responsibilities have increased; use Excel more frequently; track containers; analyse cost of sales and revenue; more volume of transactions and data	Rising analytics demands in day-to-day accounting work	Evolving role expectations in a data-driven profession
Summarise large datasets; pivot table gives report within 5 seconds; track payroll for 3,000 staff; analyse revenue and costs; identify inconsistencies	Operational use of analytics for reporting, control, and decision support	Application of analytics in professional practice
Helps to ensure accuracy; free of errors; meet deadline; less manual; make the right decision	Analytics as a tool for accuracy, efficiency, and decision quality	Application of analytics in professional practice
Fairly confident; average; somewhat confident; confidence not really that strong; still in the learning process; confident in basic Excel but not deep into advanced analytics	Moderate and uneven confidence in analytics-related tasks	Perceived readiness and capability gaps
Automation gives me 99.9% assurance; confidence increases with repetition and tool mastery	Confidence builds through practice and familiarity	Perceived readiness and capability gaps
Learned on the job; practical work experience required; reporting needs drive Excel and analytics learning	Workplace experience as primary route to skill acquisition	Skill formation pathways
YouTube; Udemy; Coursera; CFI; ChatGPT; Copilot; self-learning played a major role	Self-directed learning as a major driver of analytics development	Skill formation pathways

First-order concepts	Second-order themes	Aggregate dimensions
Self-learning alone cannot do it; need a mentor; training classes; financial modelling courses; Power BI course	Blended learning through self-study, mentoring, and professional training	Skill formation pathways
University gave theory; not practical; laid a foundation; not deep into data analytics; education system is outdated; university mostly theory and rushed	Formal education as foundational but insufficient for practical analytics readiness	Misalignment between formal education and workplace demands
Did not prepare me to interpret figures for others; no practical exposure; did not teach analysis skills	Gap between academic preparation and workplace application	Misalignment between formal education and workplace demands
Organisation sponsors training; Power BI course sponsored; training required in role; Excel mastery training provided	Organisational support strengthens analytics development	Organisational enablers and constraints
Organisation does not support; Nigeria not where they invest; support depends on employer	Uneven organisational investment in analytics capability	Organisational enablers and constraints
Forget functions when not used often; limited software access; time constraints; system and internet barriers; deadlines	Practical barriers in learning and applying analytics tools	Organisational enablers and constraints

Source: Authors' qualitative analysis of interview data (2026).

4.2.1 Overview of Participants

The six participants had different accounting and finance positions in manufacturing, banking and fintech, automobile and logistics, export and shipping, and printing industries respectively. Their career experience was between two and more than six years. The selection of participants from diverse professions was deliberate, which was inline with the purposeful sampling logic in the Gioia methodology. It helped in gathering data from different perspectives of professionals' experiences.

Participant P2, who was a payroll analyst in banking and fintech, was one of the most experienced people with regard to analytical skills in the team. He had been working in analysis oriented positions for more than six years. Participant P3 was an internal control officer in automobile industry whose position encompassed both compliance and performance analysis functions. She had sufficient organizational training support for her development. Participant P4 was a general ledger accountant in manufacturing industry who had an extensive discussion about how pivot table analysis changed his reporting

routine. Participant P5 was an accountant in export and shipping industry where she experienced no organizational support at all. Participant P1 was a general accountant in manufacturing industry whose experience included the gradual shift of roles and tasks. Participant P6 was a finance assistant in printing industry whose career was still young and therefore was confident with his limited knowledge in analytics mainly through Excel reporting.

4.2.2 Evolving Role Expectations in a Data-Driven Profession

The one theme that emerged from all six interviews with a remarkable degree of consistency was that of the transformation that is taking place within the role of an accountant, evolving from being a record keeper to becoming a contributing analyst and adviser to the management team. P3 highlighted this trend in the context of internal controls in the following way:

> *"We are no longer just process people... we work with management and advise them... we now see ourselves as finance analysts. We tend to see the point of the business owner, trying to measure strengths, trying to measure weaknesses, measure our choices and opportunities and see how to gain advantage and advise appropriately too."* (P3)

This sort of statement is typical of an evolution not only in one's career but also in one's self-identity as a member of the accounting profession. The transformation is not only that of activity but also in the perception of oneself as a professional in relation to the organization that one belongs to. The evolution from process-oriented individuals to financial analysts and consultants has its own implications for the body of knowledge possessed by contemporary accountants.

P2 offered a similarly expansive account of role evolution, grounded in a six-year career that had progressively demanded greater analytical depth:

> *"My role across six years in banking and fintech has focused more on analytics and technology-driven responsibilities... positioned me for strategic decision making. Every position has to do with making use of data, analysing data for strategic decision-making processes of the organisation."* (P2)

P1 described role change in more gradual terms, emphasising the cumulative expansion of responsibilities over time:

> *"My role has gradually expanded... I get to learn more on the job and do more than I used to when I first started."* (P1)

Even P6, who was in the earliest stage of her career and had a circumscribed analytics role in her current position, demonstrated clear awareness of the professional trajectory:

> *"It is going to go beyond bookkeeping... much focus is going to be on analytical thinking and drawing reasonable business decisions from the numbers."* (P6)

This understanding of the importance of analytics being part of the profession from P6, who had less than two years of experience, to P2, who had more than six years of experience, shows that the transformation of roles reported by research (Burns and Baldvinsdottir, 2005; Granlund and Lukka, 1998) was not just an aspiration but a reality among accounting professionals in Nigeria. The issue, which will be discussed further through the remaining themes, is whether professionals have the capability to deal with these challenges.

4.2.3 Application of Analytics in Professional Practice

Although all the interviewees reported the increasing relevance of analytics to their jobs, the type of analytics that they engaged in was not sophisticated but rather operational in nature. The most important software was Microsoft Excel, and the activities included reconciliation, report generation, performance tracking, and supporting operational decision-making as opposed to strategic decision-making.

P4 offered a particularly vivid illustration of how pivot table analysis had transformed the efficiency and accuracy of routine reporting:

> *"I use the pivot table which helped me give this report within 5 seconds... it gives an accurate report compared to manual reporting. I meet deadlines, I meet deadline in my reporting and also it gives an accurate report."* (P4)

More than the efficiency, the relevance of this narrative lies in the notion of increased accuracy, in P4's assessment of being "virtually perfect." In other words, it is an instance of analytics that can transform what professional practice entails, by eliminating concerns regarding error and substituting them with confidence in the outcomes. This, despite being relevant to the analysis of data, cannot be termed modelling or prediction analytics.

P2 described analytics use in a higher-volume and higher-stakes context, managing the payroll of over 3,000 employees:

> *"I handle the payroll of over 3,000 staff... you need all the tools necessary and you need to be accurate enough to ensure that there is no overpayment, there is no underpayment, you are paying to the right person."* (P2)

This is an example that demonstrates the use of analytics tools to deal with complexity and risks in operational accounting settings rather than simply producing reports.

The high stakes involved in payroll processes mean that there is strong incentive for organisations to utilise analytics tools that extend far beyond mere efficiency.

P3 discussed a more strategic form of analytics where analysis is done on revenues, expenses, and overheads, allowing non-financial managers to understand their business operations better:

> *"We analyse cost of sales and revenue... they only see revenue but we show them cost... we help them to see that yes, you are making money, but at the*

same time, sometimes you are also eating into your revenue because of your costs." (P3)

P5, operating in a less sophisticated analytical environment, described using Excel to track operational volumes as a basis for performance assessment:

> *"We use Microsoft Excel to track the number of containers we do per month in order to determine if we are increasing our exports."* (P5)

Taken together, these examples demonstrate some of the types of analysis that are being done by accountants in Nigeria, ranging from simple operations monitoring (P5), improved accuracy through automation (P4), and risk management in the complicated area of payroll (P2), to performance management and cost optimization (P3). All of these are legitimate forms of analysis, but all of them lie on the operational/descriptive side of the analytics continuum, as discussed in Section 4.2.2.

4.2.4 Perceived Readiness and Capability Gaps

Although the survey results suggested that the majority of the participants were prepared to varying degrees to handle their tasks, the interviews made it clear that such preparation is always contingent, task-related, and sometimes fragile. The participants were confident in their abilities in tasks that they have performed before, but unsure or even unconfident in handling tasks outside of their regular routine.

Participant P4 explained the connection between practice and confidence in the best way possible, stating that automation had removed an element of uncertainty that once hindered their confidence:

> *"It gives me this confidence that what I am doing is 100% accurate... I have a 99.9% assurance that my work is free of errors because everything is automated. What I just have to do is to refresh my pivot table and include new data."* (P4)

This example is noteworthy for two reasons. First, the confidence expressed refers specifically to the workflow pivot table reports and not to the general ability to perform analytics. Second, the basis of confidence is not proficiency but the accuracy of the automated process, there is confidence both in the professional and in the tool itself. It seems that confidence based on process automation may prove to be less stable than confidence founded on analytical knowledge.

P6 exhibited a similar pattern of confidence linked specifically to the tasks performed and limited experience with more challenging analytical tasks:

> *"I feel quite confident because I do not do too much... it is just the Excel part that I am familiar with. I am not really deep into the use of data analysis."* (P6)

'I do not do too much' is an especially telling statement, as it shows how P6's confidence is restrained by the limitations imposed by the requirements of her present position. Her

confidence is genuine, but it does have certain boundaries within which she has never really put herself to test when it comes to the greater challenges of analytics work.

P3 was the one who gave the most revealing description of the restrictions on their readiness:

> *"Sometimes when it comes like that, I was like, where do I start from? How do I get my source document? How do I get my data? My confidence level is not really that strong."* (P3)

P2 described a similar developmental position, framing readiness as a work in progress rather than an established state:

> *"I would say fairly confident... for now I am still in the learning process."* (P2)

In sum, these interviews highlight the dynamic nature of readiness that is unevenly dispersed amongst different types of tasks and professional environments, as well as influenced by dimensions that cannot be measured through surveys. This division between confidence in known tasks and ambiguity in unknown tasks, the cornerstone of the conditional readiness concept described in Chapter 5, can be observed among several interviewees and represents one of the most theoretically relevant insights from the qualitative analysis.

4.2.5 Skill Formation Pathways

One clear trend emerging from all six interviews is the reliance on self-learning and practical experience to gain knowledge of data analytics. This trend was consistent across all career levels and different organizational settings.

P4 made it clear where he learned how to perform data analysis:

> *"We actually learned these skills on the job, which is required on the job."* (P4)

P1 offered a similarly direct account, attributing analytical capability to practical workplace engagement:

> *"Most of my skills were developed through practical work experience and using the tools."* (P1)

P5 described a particularly self-reliant pathway, driven primarily by online resources in the absence of institutional support:

> *"I learned online, YouTube... self-learning played a major role."* (P5)

P6 described how personal curiosity had driven self-directed skill development alongside minimal workplace training:

> *"Out of curiosity, I had to do some practice on my own to see how I can be more confident and competent in using Excel."* (P6)

P3 described a more blended pathway that combined online tools with AI assistance, an analytically inclined supervisor, and organisational training:

> *"I go online, I use some little AI... ChatGPT, Copilot... and I also learned from my analytical boss. My organisation also put us in some trainings."* (P3)

P2 offered the most nuanced account of the limits of self-directed learning, identifying mentorship as a critical complement:

> *"Self-learning alone cannot do it. You need a mentor, you need someone that is already there or someone that actually has the understanding... I think it is 50-50. You need someone to guide you so you know that OK, I am doing the right thing. With self-learning it is like you are trying trial and error and before you get it, it takes a longer time."* (P2)

P4 described a concrete example of applied self-learning that illustrates how informal learning can produce significant productivity improvements:

> *"There was a particular task that was manually computed and it took a whole long hour. When I joined the role, I was like, oh, this is a whole lot. So from what I learned, I had to use ChatGPT, I had to search online... to automate or make the process easier. So after that it was something that took maybe an hour, took like 15 minutes to actually process."* (P2)

In this case, the effectiveness of self-directed learning is especially evident since its application in solving a concrete problem in the work environment has had an actual and demonstrable impact on productivity. At the same time, it draws attention to the fact that self-directed analytics learning is often reactionary in nature; the reason for developing it was the emergence of a concrete necessity. This means that while this type of learning creates concrete skills, it may overlook other aspects of analytics ability.

4.2.6 Misalignment Between Formal Education and Workplace Demands

An extremely consistent and prominent theme running through all six interviews concerned the perceived disparity between the information provided by formal university education regarding accounting knowledge and the requirements of the modern workplace in relation to analytics knowledge. All the interviewees that have received university-level education mentioned this issue, pointing out that university-level education was purely theoretical.

The participant P4 expressed this point quite clearly:

> *"My university education actually prepared me for the theory aspect, not the practical aspect... it does not prepare me to interpret my figures to others... it does not give me that skill to show what I actually understand in figures and in expression."* (P4)

This explanation brings out an aspect of the gap between education and practice which is more than merely technical skills; the ability to convey the results of analysis to laypeople

is becoming a vital professional skill which is not currently being taught in any educational program for accountants.

P1 presented a similar explanation, defining the gap between preparation and competence as follows:

> *"My university education provided a strong theoretical foundation in accounting and finance, but with my daily activities I get to know more about the practical aspects of it."* (P1)

P2, who had also undertaken professional certifications, characterised the role of his university education as preparatory rather than sufficient:

> *"It laid a foundation, but it did not actually go deep into data analytics. It was after university that I enrolled in financial modelling classes and data analytical classes."* (P2)

P3 was particularly emphatic about the limitations of formal educational preparation:

> *"University is mostly theory and it is rushed... it is in the curriculum, but it is not fully grounded. They do not give us the tip of the iceberg."* (P3)

P5 extended this critique beyond individual experience to a systemic assessment:

> *"The system is outdated... while in school it was basically theoretical. As a student, if you really want to get skilled, you are going to have to learn on your own."* (P5)

P6 reinforced this pattern from an early-career perspective:

> *"My university only gave me the technical knowledge... the practical experience really made me more exposed to the analysis thing."* (P6)

The recurrence of this theme among interviewees from various positions, industries, levels of experience, and academic qualifications is remarkable. This reflects a unified professional judgment regarding the sufficiency of formal training in accounting for the analytical requirements of current practice, a conclusion that is clear-cut in its nature although not necessarily consistent in its degree among all interviewees.

4.2.7 Organisational Enablers and Constraints

The last composite component considers the effect of organisational context on the analytical proficiency of individuals. From the interviews conducted, there was found to be significant differences among all six interviewees regarding the amount of support they received from the organisation for the development of their analytics capabilities.

P2 explained his experiences of working in an organisational context that invested in his analytical skills development by providing him with various forms of training.

> *"Most of these trainings I attend are being sponsored by my organisation... they encourage continuous professional development through trainings and knowledge-sharing sessions so that I can improve my skills."* (P2)

P4 similarly described an organisation that mandates analytics training as part of professional development requirements:

> *"My current role requires training in data analysis and financial modelling... my organisation is looking into and enhancing this for all finance and accounting persons."* (P4)

P3 offered an account of an organisation at the frontier of analytics investment, with active engagement in AI-focused training:

> *"My organisation is very keen on training and skill development... I am presently on an AI programming training that runs for about three months. They have given us accounting and AI training and the likes."* (P3)

P6 described a more modest but nonetheless concrete form of organisational support:

> *"The organisation provided a section... Excel mastery training... to step up your Excel skills and make your work more efficient. Not just how to use Excel, but to step up your skills so that with the help of this training, it will help you get better at what you do."* (P6)

By contrast, P5 described a context of near-total organisational disengagement from analytics development in the Nigerian context:

> *"No, they do not really support analytics skill development here in Nigeria. My organisation did not really see Nigeria as a country where they can invest... they probably invest in employees in the UK, India, Singapore and Dubai, but not here in Nigeria."* (P5)

This example is especially telling since it illustrates something more than mere resource limitations; in this case, the lack of organisational support indicates an intentional geographic investment choice, where the Nigerian labour force ranks lower than labourers from other areas. It is thus interesting to consider the effects on equity when organisations invest in analytical tools, as will be further discussed in Chapter 6.

In terms of infrastructural issues, these were brought to light by P3 through the mention of constraints outside the organisational setting:

> *"Doing proper analytics needs a very good system and also internet is a very serious barrier. Before you load a particular thing you have to go offline... I do not have my mail on my phone, my phone cannot connect Outlook, I have to use my laptop. So those gadget problems are a barrier."* (P3)

P4 identified a subtler but equally consequential barrier the erosion of skills through infrequent use:

> *"If I do not use these functions frequently, I tend to forget some key function and shortcut... we tend not to use this frequently until there is a necessity for reporting. So those are the challenges."* (P4)

P6 echoed this pattern:

> *"If I do not use it often, I tend to forget... it works better where you use data analysis frequently and regularly."* (P6)

The narratives of the loss of skills due to lack of utilization illustrate a key relationship that readiness metrics do not capture: the notion of readiness as something that needs continuous effort through consistent practice. This means that there are significant repercussions for the way organizations conduct their analytics initiatives and sustain the benefits they derive from training.

4.2.8 Summary of Qualitative Findings

Qualitative insights point to a profession wherein analytics expectations are real, increasingly present, and transformational in their effects. In practice, analytics is indeed performed, especially by means of Excel tools, and in areas such as reporting, reconciling, assessing, analysing costs, and assisting managers in decision-making. Readiness, meanwhile, is always contingent upon certain conditions; is linked directly to specific tasks; and is unevenly distributed across the profession. Confidence in analytics is high for well-known and frequently performed analytical tasks but lacking in other cases.

Learning analytics skills occurs primarily through self-study and on-the-job learning. Formal university-level education is viewed as sufficient in terms of theoretical knowledge but inadequate in addressing the practical requirements of modern accounting practice. Organisational support, moreover, is highly variable and influences the extent and nature of analytics skill development. Infrastructural issues, including internet availability, computer quality, and software access, provide further obstacles that impede analytics practice for accountants operating under resource-scarce conditions.

4.3 Integration of Quantitative and Qualitative Findings

In this case, both the quantitative and qualitative findings are highly complementary, and the integration of the two strands results in a more comprehensive and nuanced description of analytics readiness than could have been achieved through each one individually.

Firstly, there is a striking similarity in terms of skill acquisition pathways. While 61.6% of the survey participants stated that they have acquired their analytics skills through self-study and on-the-job learning processes, all the six interviews revealed similar insights about learning analytics via YouTube videos, online courses, AI tools, trial and error at work, and mentoring. Therefore, it can be stated confidently that informal learning is the key means of developing analytical skills among Nigerian accountants.

Another significant point of overlap relates to the connection between organizational support and analytics readiness. In particular, the chi square association between these two factors ($V = 0.41$) found in the survey data is given substance in the interviews, where

P2 and P4 describe situations when organizational support has led to mandatory skills development, whereas P5 clearly explains how organizational disengagement in terms of analytics learning has become an issue.

The educational background analysis is another instance of strong convergence between the strands: the survey reveals that 49.6% of the participants believe that their education has prepared them sufficiently for analytics tasks, and a chi square association between readiness and perception of educational preparation ($V = 0.48$). However, in terms of interviews, the participants unanimously admit that theory behind analytics is strong, yet the practical part is weak.

There is one instance of significant divergence between the two strands of research worth mentioning. According to the survey data, 61.6% of respondents are ready to perform analytics tasks which sounds positively but might create the wrong impression that the majority of professionals in the field is prepared for data-driven work. Nevertheless, in terms of the interview data, all the six participants provide somewhat cautious assessments of their ability to perform analyses. For example, P6 admits that she is reasonably confident because she does not do analytics very often; P2 feels confident but is in the process of learning; P3 believes that he is relatively confident in handling analytics but in unfamiliar situations; and P4 is confident only in terms of automation of routine processes.

The above divergence is theoretically relevant. In fact, the aggregated professional assessments (such as in surveys) may systematically reflect a false sense of confidence due to the limited scope and type of analytical tasks. This limitation makes it clear why a combination of survey and qualitative data leads to more valid and comprehensive conclusions.

4.4 Concluding Summary of Chapter 4

The findings of this chapter have been presented, discussed and synthesised above. Some of the key findings can be summarised as follows: there is high reliance on advanced Excel for analytics purposes, with more sophisticated tools underused; the respondents' perception regarding their readiness to engage with analytics is quite favourable overall but conditional, task-specific and highly heterogeneous; learning by self and on the job is by far the most common mode of acquisition of the relevant skills, due to lack of sufficient formal training provisions; formal education at university level is believed to be theoretically adequate but not practically so for current requirements in analytics; organisational support, educational preparedness and confidence in analytics performance are the main factors affecting readiness, while years of experience, job role and level of education make no significant difference; and lastly, structural constraints like software availability and time pressure act as barriers to analytics capability development.

Overall, these findings confirm that accountants in Nigeria are only partially and selectively ready for the era of data-driven decision making. Professional motivation and personal drive are abundant; however, lacking are the organisational infrastructure and professional development opportunities to turn that motivation into tangible analytics capabilities. Interpretation of these findings is provided in the next chapter.

4.5 Supplementary Quantitative Analysis

4.5.1 Industry and Qualification Effects

Apart from the chi square relationships outlined in Section 4.2.7, further analysis focused on exploring whether industry sector and highest qualification played a role in determining readiness. As can be seen in the data below, the industrial sector variable was just shy of reaching traditional standards of significance and consequently did not reach significance at the 5% level, chi square (8, N = 125) = 14.89, p = .061, Cramer's V = 0.24. The close proximity to significance suggests that the industrial sector may influence readiness in that those in the financial services industry may have better access to analytics tools because their organizations allocate greater resources towards analytics. However, the small expected counts make it difficult to interpret the findings without any caution.

Furthermore, the findings revealed no significant relationship between highest qualification and readiness, chi square (6, N = 125) = 3.95, p = .684. It is worth noting that this finding reinforces the general theme of insignificant background variables and demonstrates that having attained a certain qualification level does not automatically translate to an increased state of readiness when it comes to analytics abilities. From the point of view of both the professional practice and the academic discipline, it is a very important discovery since it means that the curriculum, not the qualification itself, is key to readiness.

4.5.2 Cross-Tabulation of Skill Acquisition Pathway and Organisational Support

According to the findings from exploratory cross tabulation analysis, a unique trend where individuals whose primary source of skill acquisition was structured professional training had higher chances to express satisfaction with their organisation’s provision of support for analytics relative to individuals whose primary sources of skill acquisition were self-learning and on-the-job learning, respectively (72.3% vs. 53.2%) was observed. In essence, the above-mentioned relationship demonstrates that there is a lack of independence between structured training and organisational support, but the factors appear to be highly related. Companies that provide structured training to their staff appear to be supportive in several different areas, while companies that depend on individual efforts of employees in developing their analytics skills have poor records when it comes to any kind of organisational support.

It means that the significance of chi-square association found for the relationship between the respondents’ readiness for data analytics and either structured training or organisational support cannot be interpreted independently since the two constructs cannot be seen as independent predictors of readiness, but correlated characteristics of organisational environment. Indeed, qualitative evidence confirms this point of view: the organisations belonging to P2, P3, and P4 respondents were described as supportive in several different ways, providing funded training programs, access to software tools, and opportunities for peer learning.

4.5.3 Confidence and Preparedness: Examining the Relationship

It is significant to investigate the directionality of the association since there exists a strong chi-square correlation (Cramer's V = 0.44) between confidence in use of data for decision-making purposes and perception of being ready for such use. Does confidence cause readiness? Or vice versa? Since the study was done on a cross-sectional basis, no causal relationship may be established. Nevertheless, the qualitative findings of the study shed light on the question. Confidence is built when professionals become able to automatize the process of performing certain operations (P4), or when they repeatedly engage into specific activities and thus gain enough experience to become confident in their performance (P6). However, P2's confidence-building story is interesting as it involves taking specialized training courses to build up confidence. Thus, it appears that confidence and readiness are two reciprocal notions, supporting each other in a virtuous or vicious circle depending on the environment of the organization (Bandura, 1986, p. 391).

4.6 Participant Profiles and Cross-Case Analysis

4.6.1 Participant Overview

In order to put the findings of the qualitative study into context, Table 4.8 outlines below presents a brief description of all six individuals who participated in the interviews, outlining their respective professional role, industry, experience level, main skills training route, organizational environment of training and confidence levels.

Table 4.8. Interview participant profiles.

Code	Role	Industry	Experience	Primary learning pathway	Org. support	Self-described confidence
P1	General Accountant	Manufacturing	~3 years	On-the-job	Moderate	Fairly confident
P2	Payroll Analyst	Banking / Fintech	6+ years	Self-learning + mentorship	Strong (sponsored training)	Fairly confident; still developing
P3	Internal Control Officer	Automobile	~4 years	Online / AI / supervisor	Strong (AI training)	Moderate; uncertain in new tasks
P4	General Ledger Accountant	Manufacturing	~5 years	On-the-job + Udemy / CFI	Strong (mandatory training)	High in specific routine tasks

P5	Accountant	Export / Shipping	~2 years	YouTube / self-directed	None (org. disengaged)	Average
P6	Finance Assistant	Printing	<2 years	On-the-job + self-practice	Modest (Excel training)	Confident in Excel basics only

4.6.2 Cross-Case Comparison of Key Themes

Comparing the six participants, a few themes emerge, which can be used to shed light on the quantitative results obtained through the chi square test. First, organisational support has a clear effect on the scope of analytics capability development. Three of the four participants who work in organisations that heavily invest in analytics (P2, P3, and P4) demonstrate the highest variety of analytics activities, including payroll analytics (P2), non-financial performance analysis (P3), and automated reconciliation reports (P4). In turn, the participant who works in an organisation without any investment in analytics development (P5) is engaged in the least complex activity, namely, Excel based container tracking. The participant with modest organisational support (P6) falls into the middle.

Second, the relationship between career experience and readiness cannot be described through a simple linear progression. The oldest participant (P2) with over six years of experience describes himself/herself as still being in the learning phase and actively engaged in acquiring skills by attending external training courses. At the same time, the second oldest participant (P4) with about five years of experience demonstrates very high task specific readiness. P6 with less than two years of experience demonstrates the level of baseline Excel based analysis capability. This suggests that it is rather the organisational setting as well as the type of analytics demand in the specific role, which affect the progress toward readiness rather than the stage in one's career. Thus, the insignificant chi square test association between these factors is confirmed.

Finally, the educational misalignment phenomenon is common for all five participants with university degrees. There is no example in this sample when participants felt that their education had adequately prepared them for the practical challenges related to analytics they encounter in their work. Considering this phenomenon across five participants who graduated from various universities and different specialisation degree programmes, one can conclude that this is a systemic problem rather than an idiosyncratic situation associated with any particular school or specialisation programme.

4.6.3 Negative Cases and Deviant Accounts

When performing qualitative research following the Gioia methodology, it is necessary not only to focus on the dominant theme but also to pay attention to negative cases, or participants' perspectives that diverge from or explain the dominant pattern (Gioia et al., 2013, p. 20). A particularly interesting case refers to the relationship between organisational support and readiness. Whereas the dominant pattern indicates that a high level of organisational support contributes to readiness, the interview with P3 demonstrates a more complex relationship. Even though P3 works for an organisation

that invests heavily into analytics, the interviewee shows relatively low levels of confidence when facing new analytical tasks, admitting the lack of understanding of where to begin and how to find appropriate data. It explains that, in spite of having strong organisational support, people can still experience difficulties in developing their analytics skills depending on the nature of the task they face. Thus, readiness is conditional by default rather than the result of the lack of organisational support.

Another relevant point to consider is the fact that P5 manages to develop some skills despite not having organisational support. Working in a less favourable environment structurally impedes individuals' ability to learn something; however, P5 shows real interest in analytics, thus developing certain abilities through self-teaching via YouTube videos.

4.7 The Future of Analytics in Nigerian Accounting: Participant Perspectives

Apart from problems associated with their current experience and practices, another topic was discussed among all six participants regarding the skills that accountants will need in the next five to ten years. This information about future accounts is vital in understanding how prepared they think accountants currently are and also sheds light on their perspectives concerning the development of the field.

In addition, all six interviewees agreed that analytical skills will become increasingly important, and they all shared the idea that accounting will have to develop far more than it traditionally does, focusing solely on accounting entries and reporting. In particular, P4 presented his perspective on accountants becoming analysts and interpreters instead of just recorders:

> *"As an accountant, we need to move beyond just giving figures, recording figures, and we need to move beyond that and start presenting our figures and reporting our figures in a proper way." (P4)*

P2 emphasised the importance of automation and AI tools in the future of accounting work:

> *"The world is moving into automation and data analytical approaches where you can make decisions for the organisation. You are not just an accountant that would just do bookkeeping. You have to start analysing and making decisions for the organisation." (P2)*

P3 was particularly emphatic about the role of AI in the profession's future:

> *"AI is not coming to replace anybody. AI is just coming to make life better and make work easier. If as an accountant you are seeing AI as the enemy, very soon you will be out of a job. You need to embrace artificial intelligence into whatever it is you are doing." (P3)*

P5 framed the future challenge in terms of the distinctive judgement that accountants bring to data analysis:

> *"As an accountant, yes, we need data analytics skills, and then we also need decision-making skills being able to make accurate decisions after*

compiling all the data, tracking performance and using your knowledge and experience to make decisions. That is what is going to set us apart." (P5)

P6 articulated the profession's transition with particular clarity:

"It is going to go beyond bookkeeping... much focus is going to be on analytical thinking and drawing reasonable business decisions from the numbers." (P6)

There are several reasons why these speculative analyses are important. Firstly, they show that the question of analytics readiness is not only relevant at this point in time but is also something that is going to be even more relevant in the coming ten years. Secondly, it shows that the people who are involved are fully aware of this trend and are thinking about how they can get ready for their future career. If this is effectively implemented via training, this may prove to be a very important catalyst for the development of analytics skills among accountants in Nigeria.

4.8 Additional Methodological Notes on Data Integration

The mixed methods convergent design adopted for this study can be described as one among many alternatives that can be chosen when attempting to integrate quantitative and qualitative information into the same study. Some of the alternative designs that could have been considered but eventually rejected include an explanatory sequential design, where quantitative data collection and analysis would precede and help guide qualitative data collection and analysis, and the exploratory sequential design, where qualitative data analysis would help in designing a quantitative tool. The convergent design was preferred since both qualitative and quantitative strands were based on an established theory, which made it unnecessary to use one strand to develop another one. Simultaneous analysis of both strands enabled their completion in the available time frame.

Integration of quantitative and qualitative findings in this study has been carried out by utilizing the joint display technique where the quantitative results have been compared to the qualitative results by placing them together and conducting a comparative analysis of the findings. To achieve the goal, the research involved developing a comparison table in which each result derived from the quantitative analysis had been compared to pertinent qualitative themes in order to establish whether there was convergence, divergence or which strand provided rich context.

The concept of triangulation becomes very important when evaluating the quality of integration in this research design. In mixed methods research, the use of triangulation does not mean that the researcher is expecting similar findings from different methods but rather employing a combination of methods to provide a comprehensive explanation of the phenomenon in question (Bryman, 2016, p. 394). Divergence of qualitative and quantitative findings in this case shows how the use of the mixed methods research design works effectively by helping in explaining a complex nature of the phenomenon (conditionality of readiness).

5. Analysis and Discussion

This chapter provide a comprehensive analysis and discussion of the empirical findings presented in Chapter 4 under the theoretical constructs of role theory, human capital theory, and professional competence theory outlined in Chapter 2. This chapter is not about restating empirical findings that have been presented in Chapter 4 but exploring their theoretical implications: identifying conflicts and contradictions both within and between the two strands of evidence; placing empirical findings within the existing body of scholarly discussions; and articulating a conceptual contribution that the study makes to academic knowledge in the area of data analytics readiness in accounting practice.

The layout of this chapter will be based on the conceptual framework provided in Section 2.7, starting with each of the three lenses and ending up with the synthesis of these theoretical constructs. Specifically, Section 5.2 discusses the empirical findings through the perspective of role theory, considering issues of role transition, role conflict, and role ambiguity in the context of professional transformation driven by the emerging data analytics technology. Section 5.3 applies human capital theory to examine the issue of skill acquisition, looking into reasons for the predominance of informal learning approaches to developing analytics skills within the profession. Section 5.4 employs professional competence theory as a theoretical tool for interpreting the findings regarding readiness to adopt data analytics tools. It involves a theoretical introduction to the notion of conditional readiness. Section 5.5 deals with the role of organizational context as an enabler or a constraint. Finally, Section 5.6 combines all the theoretical perspectives to define the theoretical contribution of the study. Section 5.7 answers the research question directly. Throughout this chapter, when dealing with results of chi-square tests, care will be taken of the interpretive caveats outlined in Section 4.2.7.

5.1 Data Analytics and the Evolving Role of Accountants: A Role Theory Perspective

As per Biddle (1986), according to role theory, the idea of professional roles is not rigid, and it is constantly affected by institutional, technical, and organizational influences. The role change theory by Biddle gives us great insight when we are analyzing situations where the incumbent of a particular role is faced with challenges created by the environment, which he cannot control. In this situation, the conclusions drawn from this study offer evidence in support of the role change theory.

The most important finding emanating from qualitative analysis is the consistent mention of role transformation among all six participants. It must be stressed, however, that irrespective of the industry the individual comes from, the experience he possesses, or his rank within the organisation, all participants mentioned that their role was undergoing transformation in the same manner, shifting from transactional to advisory. Specifically, P3 stated that role transformation entails:

> *"We are no longer just process people... we work with management and advise them... we now see ourselves as finance analysts."* (P3)

The reason why the statement 'we now see ourselves as finance analysts' carries theoretical importance is because of how it addresses both identity and function. According to Burns and Baldvinsdottir (2005, p. 726-742), role theory has demonstrated that role transitions occur not only in terms of altered task profiles but also in relation to

self-presentation within organisations for accounting professionals. The transition from mere process-following to strategic advice through analytics represents a radical shift in the positioning of accounting professionals within the organisational hierarchy. New competence needs, altered interpersonal relations, and new legitimacies are part and parcel of such a change of positioning.

Role transitions have been described by P2 in terms of a six-year career in banking and fintech that gradually became dominated by analytical practices. The field of financial services, in Nigeria at least, has always been one of the data intensive fields within the economy. P2's description of a career path 'focused more on analytics and technology-driven responsibilities' which prepares the way for 'strategic decision-making' demonstrates the degree to which this sector represents a locus of role transformation. In addition, managing payroll data for more than 3,000 employees using analytics shows how deeply ingrained analytics has become in operational processes.

It should be noted here that the survey findings offer an empirical foundation for the above-mentioned qualitative insights. The fact that 51.2 % of the sample use advanced Excel in a very frequent fashion clearly shows that advanced analytics has entered professional practice in terms of everyday operations. However, the finding that 47.2 % of the sample uses Power BI and Tableau which are advanced analytics tools at a low frequency means that the actual analytical activities undertaken by the sample lag behind the role expectations expressed by the qualitative participants. This discrepancy was exactly predicted by role theory's account of role transition.

Biddle (1986, p. 71-84) differentiated between role conflict where individuals feel contradictions in expectations set out by different stakeholders and role ambiguity which means that individuals find themselves in an uncertainty situation. The latter dynamic is best reflected in P3's statement regarding role ambiguity in its purely analytical form:

> *"Sometimes when it comes like that, I was like, where do I start from? How do I get my source document? How do I get my data? My confidence level is not really that strong."* (P3)

This is not a case of low motivation or poor interest in analysis. P3 has a wealth of experience in their professional work and has access to organisational training, as well as an analytic-oriented boss. The feeling of uncertainty mentioned above has a deeper structure because the new role requires analytical input where no analytical frameworks and data accessibility exist.

Hopper (1980, p. 401-409) identified a mismatch between the formally described roles of accountants and the actual practice carried out by management accountants when he found that enacted professional behaviour often did not match formal role expectation. The present study highlights the modernised form of the mismatch identified by Hopper; while the expectations concerning the analytical input of accountants have grown significantly, the practice enacted by the majority of professionals is largely operational and revolves around Excel. As shown by P4's comment regarding pivot tables being the core tool in their work as a significant yet insufficient skill, this mismatch can be demonstrated particularly clearly.

As highlighted by the role theory framework, the issue of analytics readiness of Nigerian accounting professionals is a question not only of skills, but a case of structural delay; the institutional structures designed to prepare professionals for their roles lag behind the development of role expectations. This structural characterisation of the issue brings to light important implications for addressing the issue.

5.2 Development of Analytics Capability: A Human Capital Theory Perspective

Human capital theory, as formulated by Becker (1993), conceives knowledge and skills as productive investment that accumulate through education, training, and experience and result in returns in the form of professional performance, employability, and salary. Applying the theoretical framework to the case study of the development of analytics capability amongst Nigerian accountants, it is clear to see that the resulting picture reflects a unique path of human capital accumulation that is unique in its methods, its inconsistency, and implications for sustainability.

The first thing worth noting about the human capital theory when applied to this phenomenon is how overwhelmingly informal and self-directed the pathways for accumulating the required skills have been. Survey data show that the percentage of people who obtained their skills through informal channels reaches as high as 61.6 % compared to just 25.6 % of those who relied on professional training. This fact is confirmed by interviewees across the sample (P1 P6). Specific ways of obtaining skills mentioned include watching YouTube tutorials (P5 and P2), participating in courses on Udemy and CFI (P4), using AI-assisted tools such as ChatGPT and Copilot (P3), taking courses on Coursera (P2), developing the necessary capabilities through practice and experience on the job (P1 and P4), and learning from an analytically-minded boss (P3).

The above-described pattern of human capital accumulation can be explained by considering Becker's (1993) distinction between general and specific types of human capital. While a formal education at a university provides general human capital in the shape of fundamental theoretical knowledge of a certain profession, more specific skills, which would include proficiency in using a particular ERP system, producing dashboards in a format requested by management, and utilizing Excel functions relevant to a specific type of transaction, constitute specific human capital that can only be gained through practical experience on the job. Exactly this process is described by interviewees as the way they developed their analytics skills in response to specific job-related requirements using specialized resources.

Another useful addition made to the human capital framework comes from Nafukho et al. (2004, p. 545-549), who claimed that the productivity of investments made depended on the degree of relevance of the skills acquired to the current demands of one's role. The chi-square test showed that years of experience and the level of formal qualifications of the respondents are unrelated to the degree of readiness, while skill acquisition channel is highly relevant (Cramer's V = 0.30). Informal acquisition of human capital, as opposed to the time spent acquiring formal credentials and working, is what contributes most to the professional performance of the individuals interviewed.

The discussion provided by P2 on self-directed vs. mentored ways of learning is particularly informative in this respect:

> *"Self-learning alone cannot do it. You need a mentor, you need someone that is already there or someone that actually has the understanding... I think it is 50-50. You need someone to guide you so you know that OK, I am doing the right thing. With self-learning it is like you are trying trial and error and before you get it, it takes a longer time."* (P2)

The value of this account for theory lies precisely in its identification of the social context required for informal skill development to be effective. While the human capital approach traditionally focuses on the skill acquisition as a relatively solitary process, the findings of this study suggest that, at least when we consider the development of analytics skills, a key part of the equation is the formation of relational human capital where one can benefit from advice, encouragement, and guidance provided by colleagues who possess more experience in a certain area. In this regard, relational human capital is evident in the story shared by P2. However, the analysis of narratives from P3 and P6 shows that relational human capital is an integral component of the informal learning processes they describe.

The proportion of survey participants who stated that they have not developed any analytics skills whatsoever, 12.8%, needs to be highlighted in light of the human capital approach. These people represent a segment of professionals for whom the informal mechanisms of developing analytics skills available to their colleagues had not come into play. Such failure could be explained either by the absence of motivation related to their role or by a lack of necessary learning resources. For professional associations and organisations interested in developing analytics skills among their employees or members, an important question would be understanding why those individuals did not participate in self-directed learning and what sets them apart from 61.6% of survey respondents who did participate in such learning.

One major limitation of the human capital account of analytics skill development noted in Section 2.4.3 lies in its tendency to overlook the structural aspect of skill development. While the evidence of this case study suggests that the conditions of developing analytics human capital are highly influenced by organisational factors training, funding of necessary tools, allowance of professional development time, organisational culture of valuing analytics this is not considered adequately. As in the case with the mixed account of self-learning and mentored learning offered by P2, the organisational aspects of mentoring were not available to all professionals in equal measure.

5.3 Professional Competence and Readiness: Conditional Readiness as a Theoretical Contribution

According to Mulder (2014) synthesis of the professional competence theory, competence consists of the combination of knowledge and skills alongside the ability to utilise such a combination effectively in particular circumstances. One of the important features of the theoretical perspective is that competence, as the combination in question, involves subjective experience and specifically perceived readiness one's evaluation of the ability to handle role demands (Mulder, 2014). Thus, this is an important dimension of professional competence, which, at the same time, is not identical to objective skill assessment.

The empirical evidence provided by the research demonstrates a type of perceived readiness that cannot be explained by the extant conceptual frameworks of the phenomenon in question. On average, positive results on the survey-based measures indicate a certain level of readiness among professionals. However, the detailed examination of qualitative evidence suggests that it exists in a more conditional manner readiness that is strong and reliable when dealing with familiar and well-practiced analytical tasks, but that weakens and even becomes absent in cases where one encounters unusual data challenges or novel analytical demands or faces a task beyond habitual repertoire. It should be noted that the proposed pattern was not associated with any motivational or individual shortcomings and thus should not be viewed in this way. Rather, it is an inherent trait of a certain process of skill development, which relies on reactiveness, informality, and task specificity rather than systematic approach, institutions, and comprehensiveness.

The available qualitative evidence on conditional readiness is plentiful. For example, P4's discussion of confidence that rests upon trust into the process of automated generation of a pivot table reveals that it can be a powerful foundation for readiness only within the scope of this very task. Similarly, P6's clarification of her readiness as being linked to her not having too many things to do in the current role shows that, again, it exists only within a narrow scope. Finally, P3's indication of her feeling of uncertainty as she encounters an unfamiliar challenge 'where do I start from? How do I get my source document?' reveals how fragile such readiness is when developed in an ad hoc and unstructured manner.

In relation to current theoretical perspectives, the theory of conditional readiness represents both an expansion and qualification of existing approaches to professional competence. As Mulder (2014) conceptualised it, perceived readiness was seen as self-assessment of capacity to meet role demands, thus being considered as relatively stable. In the context of this research, such a view of perceived readiness should be qualified as the following: it is not stable, but situation-dependent, as it depends on the nature of the analytical task, the extent to which the professional has mastered analytical tools, and the availability of situational supports. It is for this reason that situational variations in perceived readiness are not revealed using survey items measuring perceived readiness on an aggregate level, which results in differences between survey findings and more qualified interviewees' responses about their perceived readiness.

As a related theoretical perspective explaining conditional readiness phenomenon, it is important to mention Bandura (1986, p. 391) concept of self-efficacy, an individual's self-belief in his or her capacity to successfully perform particular behaviours. According to the theory, self-efficacy can be attributed only to a particular kind of performance, since the same individual can have very high efficacy with regards to some activities but not others. From this point of view, conditional readiness implies that the professional has self-confidence in performing certain analytical tasks but lacks confidence with regard to other, presumably more complicated ones, which explains why the general measures of self-confidence yield results different from the task-specific ones obtained during interviews.

Finally, the result obtained through chi-square analysis, which indicates that perceived educational preparation serves as the strongest predictor of readiness for analytics (Cramer's $V = 0.48$), should be interpreted carefully with respect to conditional readiness. To start with, it cannot be concluded that formal education produces analytics readiness,

as the overwhelming amount of qualitative data indicates that formal education was considered as theoretically sufficient but practically inadequate for fulfilling professional role requirements. In addition, from the standpoint of Eraut (1994) theory on formal education and professional competence, professionals who perceive their formal preparation as adequate are in a better position to assimilate workplace and self-learning experiences. Therefore, educational quality can serve as a prerequisite for further capability development rather than a sufficient condition. Such an interpretation makes it clear that the association between educational preparation and perceived readiness is stronger than the association between skill acquisition pathway and readiness because education affects capabilities.

In conclusion, the presented concept of conditional readiness has important implications for further research on the subject. First, research in the field should include not aggregate but task-specific readiness measures, separating the perceptions of operational and more difficult tasks. Second, practical measures aimed at improving analysts' readiness should focus not on developing self-confidence in operational activities but in advancing skills in performing more challenging tasks. Third, the standards and framework used by professional organisations for assessing professional competence should allow for situational variability.

5.4 The Role of Organisational Support

The empirical evidence from this study consistently identifies organisational support as one of the most powerful determinants of analytics readiness, ranking third among the chi-square predictors of perceived preparedness (Cramer's V = 0.41) and running as a central theme through the qualitative accounts of all six participants. The importance of organisational support extends beyond its direct statistical association with readiness to its role as a moderating variable that shapes the quality, depth, and sustainability of individual analytics capability development.

The contrast between the organisational experiences of P2, P3, and P4 on one hand, and P5 on the other, is illuminating. P2 described an organisation that actively invests in analytics capability through sponsored training, professional development programmes, and the funding of a Power BI course currently underway. P3 described an organisation at the frontier of analytics investment, with current participation in a three-month AI programming training funded by the employer. P4 described an organisation that mandates analytics training as a formal component of the finance function's professional development requirements. Each of these descriptions reflects an organisational culture in which analytics capability is treated as a strategic asset worth investing in, and in which the development of individual professionals' analytical skills is understood as contributing to organisational effectiveness.

P5 described a radically different organisational reality:

> *"No, they do not really support analytics skill development here in Nigeria. My organisation did not really see Nigeria as a country where they can invest... they probably invest in employees in the UK, India, Singapore and Dubai, but not here in Nigeria."* (P5)

This account is remarkable for what it reveals about the intersection of organisational investment decisions and geographic inequality. The absence of organisational support

for analytics development in P5's context is not simply a resource constraint but a deliberate strategic choice to deprioritise the Nigerian workforce. This has direct and visible consequences for P5's analytics capability: in the absence of organisational support, P5 relies entirely on YouTube-based self-learning, describes an average confidence level, and works in an organisation that has not invested in the analytical software that might enable more sophisticated work.

Appelbaum et al. (2017, p. 35-41) identified organisational conditions, including access to appropriate technologies and a supportive learning environment, as critical mediators of analytics capability development among accounting professionals. The findings of the present study substantially confirm and extend this claim in the Nigerian context. Not only does organisational support predict self-reported readiness at a statistically significant level, but the qualitative accounts reveal the specific mechanisms through which support operates: structured training programmes develop systematic capability; mentorship provides relational human capital; software access enables practice; and a culture that values analytics creates the professional incentive to invest in skill development.

Lave and Wenger's (1991, p. 98) theory of situated learning and communities of practice provides a complementary account of how organisational context shapes learning. The theory holds that professional knowledge is most effectively acquired through participation in communities of practice social structures in which less experienced practitioners learn by working alongside and observing more experienced colleagues in the context of shared professional activity. P2's description of a 50-50 relationship between self-learning and mentored learning, and P3's account of learning from an analytically oriented supervisor, both reflect the situated learning dynamic: analytics capability is built not through abstract study but through embedded, relational, and context-specific professional engagement. The organisational conditions that enable communities of analytics practice including access to tools, time for collaborative problem-solving, and leadership that models analytical thinking are therefore not merely support mechanisms but constitutive conditions for the kind of learning that produces robust and sustainable readiness.

Fijabi and Lasisi (2023, p. 71-79) specifically documented the constraining impact of limited software access and training facilities on analytics usage in Nigerian accounting practice. The present study's qualitative evidence reinforces this finding at the individual level: P5's explicit identification of software access as a barrier, P3's account of internet connectivity and equipment quality as constraints, and P4's description of skill erosion through infrequent use all point to the ways in which infrastructural limitations compound the effects of insufficient organisational support.

The implications of these findings for the design of analytics capability development strategies are significant. If readiness is substantially shaped by organisational support and the evidence of this study suggests that it is then strategies focused exclusively on individual skill training will have limited impact in contexts where the organisational enabling conditions are absent. Professional bodies, employers, and policymakers who seek to close the analytics readiness gap in the Nigerian accounting profession must therefore engage organisations as active participants in capability development, rather than placing the entire burden of adaptation on individual professionals who may lack the organisational support needed to translate their motivation into capability.

5.5 Integration of Findings and Theoretical Contribution

In the foregoing discussion, the study results have been interpreted from the perspective of each of the theories role theory, human capital theory, and professional competence theory. In this final part, the discussion will synthesise the insights generated by the above analysis to highlight the contributions made by this study to the extant literature.

As indicated in the analysis, there is a convergence of insights across the three theoretical perspectives. Namely, readiness is determined by the interactions of the following factors: evolving role expectations, generating the demand for analytics capabilities (role theory); informal, cumulative, and relational learning, developing the required human capital (human capital theory); and subjective perceptions of conditional readiness, which vary based on task and situation (professional competence theory). These dynamics occur within the organisational context, which either facilitates or impedes the learning process and the implementation of readiness.

First, this study introduces the concept of conditional readiness and empirically grounds its importance as a means of understanding analytics readiness in the Nigerian accounting profession. Defined and developed in Section 5.4, conditional readiness extends the traditional professional competence theory account of readiness as a stable aggregate attribute by incorporating the notion that perceived readiness can be situationally conditional. Previous applications of the professional competence theory in the context of accounting have focused on readiness as an unchanging attribute of the individual professional's overall level of expertise. However, the data from this study suggest that the latter characterisation conceals important variability that should be taken into consideration. Conditional readiness offers a superior conceptualisation of readiness and is a more practically relevant framework than existing approaches. It has implications for readiness measurement, development, and assessment.

Second, this study extends the human capital theory framework to investigate the dynamics of analytics human capital development in the emerging economy context. The finding that informal and autonomous learning is the primary mechanism for developing analytics human capital and that neither formal credentials nor years of experience predict readiness supports Nafukho et al. (2004, p. 545-546) assertion that human capital investments pay off only when they develop skills that align with current job requirements. In the case of Nigeria, this result suggests that the human capital built via conventional accounting education is partially outdated in its analytics dimensions, making a continuous investment in analytically relevant human capital necessary.

Third, this study documents the presence of a role gap in the Nigerian accounting profession between the institutionally defined analytics expectations and the actual analytical behaviour of practitioners. The role gap, described by Biddle (1986) and Hopper (1980), is an expected characteristic during periods of rapid role change. Although the tension between evolving role expectations and enacted professional behaviour has been documented in developed economy contexts (Hopper, 1980; Burns and Baldvinsdottir, 2005), this study contributes to this literature by providing empirical evidence of the same phenomenon in the Nigerian professional context. In addition, this study adds to the knowledge base by offering a contextually specific account of the conditions, such as educational mismatch, insufficient organisational investment, and infrastructure deficits, which exacerbate the role gap in Nigeria.

Fourth, this study makes a contribution to methodology. By using the convergent mixed-methods approach, it has been possible to detect a theoretically significant divergence between aggregate readiness scores, obtained through surveys, and more nuanced self-assessments, expressed in interviews. Specifically, interview participants tended to report more conditional perceptions of their own readiness. This finding is a contribution to knowledge in itself because it reveals a potentially systematic bias associated with survey methods in measuring readiness. Future research studies should consider this finding in the design of their measurement tools.

These four contributions advance the existing literature on analytics readiness in accounting in three directions. Namely, they enhance the conceptualisation of readiness with the conditional readiness framework; offer empirical evidence on the dynamics of human capital development in the emerging economy environment; and document and explain the conditions under which the role gap dynamic occurs in Nigeria.

5.6 Answer to the Research Question

On the basis of the theoretical analysis provided in this chapter, drawing upon the theories of roles, human capital, and professional competence, it is now feasible to suggest a model for the development of analytics readiness. This will integrate both empirical and theoretical components of this study in one theoretical construct. The model proposed in this research is presented in Figure 5.1 below. The model should not be confused with the summary of the results. It is a conceptual model representing analytics readiness development in the Nigerian accounting professional environment.

There are five interrelated variables in this model. Starting from the top, evolving role expectations supported empirically from all six interviewees and theoretically through role theory (Biddle, 1986; Burns and Baldvinsdottir, 2005) create a structural need that makes analytics readiness a necessity rather than an enhancement in the accounting profession. Currently, these evolving expectations are running ahead of the available institutional support, creating an institutional readiness gap that is represented at the bottom of the model.

Analytics capability development occurs in three distinct pathways, including formal education, professional training, and self-directed and job-based learning. Each pathway is differently effective and has different accessibility levels. While formal education (purple) serves as the foundation of theoretical knowledge, it alone is not sufficient, as was consistently stated by all six interviewees. Professional training (teal) is effective yet available only to 25.6% of the study's sample as the main learning pathway. Self-directed learning (amber), available to 61.6% of participants in the sample, is more dominant yet reactive in nature, task-oriented, and varies in terms of depth and comprehensiveness. Therefore, it generates a skillset tailored to the needs of a particular situation and not comprehensive analytical skills. Organisational support is a moderating factor influencing all three learning pathways. Firstly, it increases effectiveness of training investments. Secondly, it enables the necessary practice and tool access. Finally, as suggested by interviewee P2's account, organisational mentoring relationships are required as an addition to self-study. A Cramer's V value of χ-square test performed to detect associations between organisational support and learning pathways of this type (Cramer's $V = 0.41$) proves this moderation effect.

The output of this development process is conditional readiness, the principal theoretical finding of the study. It is represented in the model as a two-zone structure with a flexible boundary between the zones. The strong zone is the set of activities associated with analytics for which professionals feel truly and confidently prepared, that is, familiar, practised, and routine activities, such as pivot table reporting, ERP reconciliation, and data tracking. The fragile zone includes activities for which professionals feel unprepared as of now complex statistics, dashboard and business intelligence, analysis of data in new environments. The boundary is movable; it shifts to the right, meaning that the strong zone grows in size due to professionals gaining experience and investing into capability development. However, given the present institutional circumstances, this boundary is positioned closer to the operational side of the analytics continuum for most members of the sample.

The institutional readiness gap illustrated at the base of the model is the main result of this study. This gap does not emerge because of unwillingness or incapability of any individual; it emerges as the mismatch between the requirements of evolving roles and skills developed by individuals as of now. Therefore, closing this gap requires coordinated actions of all involved parties across all learning pathways considered.

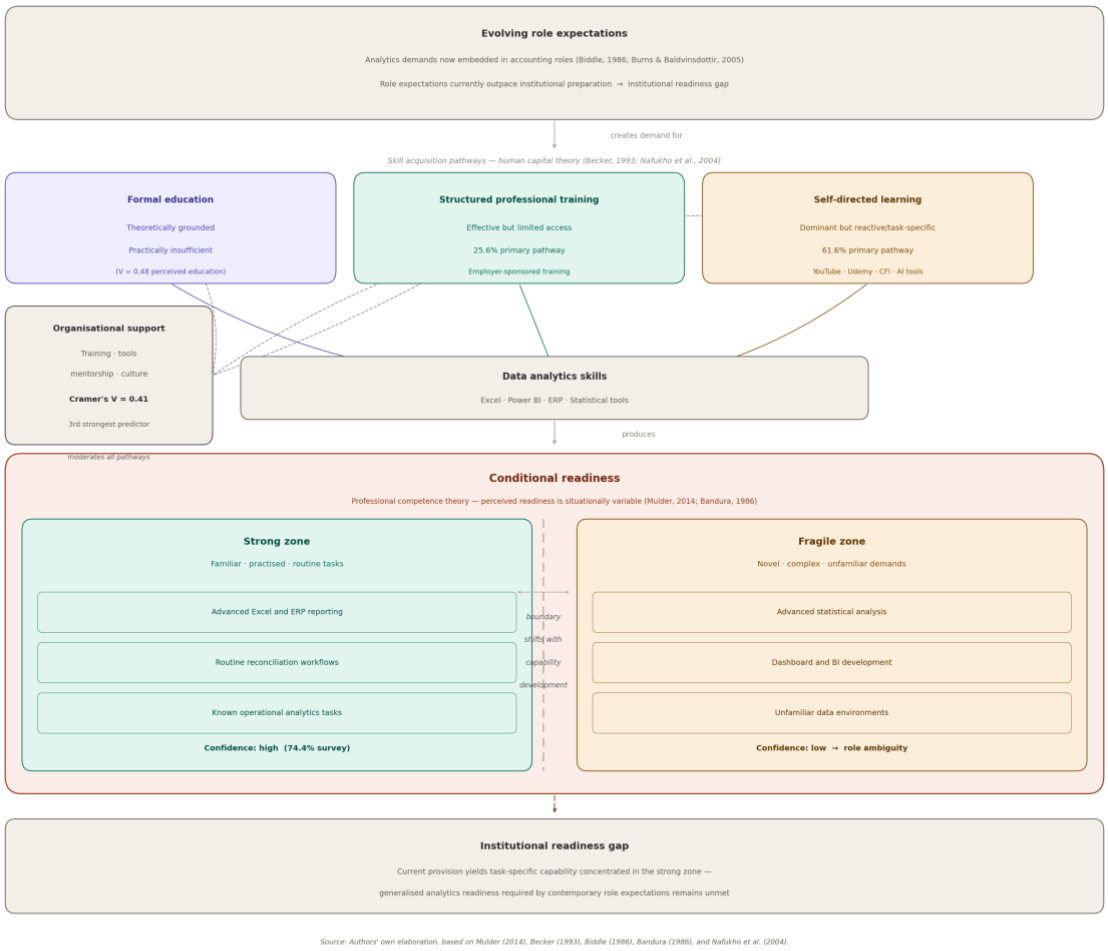


Figure 5.1. Analytics readiness development model.

The key research question posed by this study was: 'To what extent are accounting professionals working in Nigeria equipped with data analytics skills required in a data driven accounting environment?'

An answer to this question can be synthesized from the analysis of the results presented in Chapters 4 and 5. Nigerian accounting professionals are partially and conditionally equipped to perform in the data driven accounting environment but some critical gaps exist. Specifically, there are three aspects of this statement: tool usage and skills, readiness, and skill acquisition challenges. Each dimension can be discussed separately.

Firstly, regarding tool usage and applied skills, accounting professionals in Nigeria employ different analytics tools in their professional work. Advanced Excel functions, such as pivot tables, Power Query, and VLOOKUP are employed by 51.2% of the participants with fairly high frequency. ERP analytics tools are common as well. More

complex platforms like Power BI and Tableau, as well as statistical software R and Python are employed by 47.2% of respondents only to some extent. The most frequent analytical operations include reconciliation, reporting, performance tracking, payroll management, and cost analysis. While these operations can indeed be seen as analytical contribution, the problem is that they are more of an operational character and cannot be viewed as belonging to the advanced and strategic side of the analytics continuum.

Secondly, as far as readiness concerns, the majority of participants surveyed (61.6%) felt ready to undertake any analytics-related task. In addition, confidence of participants to utilise data to make managerial decisions was high 74.4%. However, empirical findings show that their perceived readiness depended on the novelty of the task. This conditional readiness is another important concept introduced into the research and constitutes one of the key theoretical findings of this study. The key drivers of readiness in a professional context proved to be perception of educational readiness (V = 0.48), confidence in analysis skills (V = 0.44), and organisational readiness support (V = 0.41). Years of experience, job role, and level of qualification were not related significantly to readiness; thus, readiness depends on capability development but does not depend on professional hierarchy and education level.

Thirdly, in regard to analytics skill acquisition, Nigerian accountants reported that their analytical competencies were mostly acquired using self-learning and gaining experience on the job (61.6%). In other words, they acquired them independently from institutions. Participants unanimously claimed that higher education in the area of accounting had provided them with a good theoretical background but lacked practical training. Moreover, all six interviewed participants agreed to the fact that their universities offered courses on accounting theories but not on analytical practices. Therefore, the main factors hindering the acquisition of analytics skills are absence of organisational training provisions, limited access to advanced analytics software tools, time pressure, depreciation of skills due to infrequent employment, and lack of infrastructural support.

In conclusion, it can be said that Nigerian accounting profession stands at an important transitional stage at the moment. Professional role evolution directions are obvious and individual professionals actively adjust themselves to the emerging needs to perform analytics-related tasks independently from organisations. Nevertheless, certain conditions for the successful completion of the process are absent. Thus, the issue needs to be addressed collectively by universities, professional organisations, and firms.

5.7 Implications for Professional Identity in Nigerian Accounting

The implications of the empirical findings of this study based on the above-discussed theoretical framework in Chapter 2 have two aspects professional capability development and professional identity. The perspective of role theory (Biddle, 1986) makes the latter clear. According to Biddle, a transition of roles inevitably implies a change of professional identity because identity is related to the role that a professional plays in the organisation. Thus, a transition from the role of 'record keeper' to 'business adviser and data analyst' implies an identity change for Nigerian accountants.

Interviewees in this study demonstrate that the professional identity of Nigerian accountants is changing. In particular, P3's statement that 'we now see ourselves as finance analysts' is the best illustration of this. P2's discussion of his career path in terms

of 'analytics and technology-driven responsibilities … [that are] positioned for strategic decision-making' suggests a different kind of professional identity. Even the most junior member of the interview group, P6, understood that accounting would change towards 'analytical thinking and drawing business decisions from the numbers'.

Despite a general transition of the professional identity, it can be observed that the process is not evenly distributed across all interviewed accountants. Those who have more junior positions in organisations that do not engage in development of analytics competencies remain at the previous stage of professional identity transition. Thus, P5 sees himself as a person working in 'organisational context where we do not develop our analytics skills'. His current identity of 'accountant' cannot evolve towards the one of 'business analyst', which makes it impossible for him to fully embrace his understanding of a Nigerian accountant as a professional whose key task is providing decision support to managers by interpreting financial information.

Such a gap between professional identity expectations of individuals and professional reality created by their organisation is likely to have a negative impact on talent retention within organisations. Moreover, it can lead to dissatisfaction of accountants who, being aware that they will perform a role of data analysts in the future, are currently engaged in purely transactional activity in the same organisation. Professional organisations (ICAN and ACCA) have an important role in this process of identity shift because they can influence perception of a chartered accountant as data analytics expert rather than just an accountant who has knowledge of certain analytic tools.

5.8 Comparison with Findings from Developed Economy Contexts

The findings obtained within the scope of the current study serve to both confirm and complement findings of the practice-focused research undertaken in developed economy contexts. Appelbaum et al. (2017, p. 42) noted that there is a gap between the expectations employers have from their employees' analytical skills in North American practice, and the actual level of competence in such analytics possessed by the professionals. The current study confirms the presence of such a gap in Nigeria and proves that it is the product of an institutional lag, not individual incapability. Similarly, Sutton et al. (2016, p. 61,68) argued for the importance of systematic studies on how analytical skills can be developed in practice. The current work provides precisely such evidence.

Nevertheless, there are aspects of the Nigerian reality uncovered during the study which cannot be explained through the findings of the developed economy practice research literature. First, the specific aspect of geographical inequality revealed in P5's story of an international employer not willing to invest in developing the analytics capabilities of its Nigerian workforce but willing to do so for its European and Asian colleagues stands out. Such unequal treatment calls into question the professional ethics of such organisations in the globalising world.

Second, the current research revealed that the process of analytics skill development in Nigeria relies predominantly on self-directed approaches. Richins et al. (2017, p. 65-74) showed that there are different paths of acquiring such skills by accounting professionals in North America. Nevertheless, their research found that structured professional development opportunities were far more common than they were in the Nigerian sample reported herein (61.6% vs. 25.6%). This disparity demonstrates that the institution

providing for the professional development of Nigeria's analysts needs improvement. Moreover, it poses some challenges to implementing the measures taken to address the analytics readiness gap since many strategies effective in developed countries may fail due to the less developed institutional infrastructure for professional development in Nigeria.

Another aspect of comparison involves the usage of AI tools to develop analytics skills. Although the usage of ChatGPT and Copilot as learning tools was discussed by many interviewees within this study, it should be noted that the usage of such technologies to assist in the development of analytics skills is not common in previous research regarding developed economies. It might be said that the usage of AI tools as learning scaffolding through ChatGPT to learn how to tackle unknown analytics problems, as explained by P2 and P3, presents a new way to learn analytics skills without extensive guidance. Future researchers should explore the impacts of AI tools on developing analytics skills and whether there are distinct outcomes between those who have learned analytics skills using AI tools and those who have learned through other methods.

5.9 The Analytics Readiness Development Model

The theoretical framework and results from the study permit the construction of a model of the process of building analytics readiness on the part of Nigerian accountants. Based on the results of both strands of this research, a model can be formulated, which has four salient properties.

First, analytics readiness grows through a reactive rather than a proactive process. According to the overwhelming majority of professionals interviewed here, analytics readiness emerges as a response to specific needs, arising in a professional's work rather than as a result of systematically prepared and managed process of analytics capability development. Such a pattern manifests itself through managers developing their analytics skills and knowledge to deliver a report that a superior asked them to produce; learning to analyse and interpret data generated by an ERP system implemented in one's organisation; and acquiring new capabilities to perform the tasks that have become too cumbersome and difficult to manage without the support of analytics tools. This reactive approach to analytics learning tends to foster task-specific readiness, which is effective enough to perform the task prompting analytics development but is not guaranteed to help deal with the more complex cases that arise from time to time. It is this logic that explains why readiness is high when it comes to performing analytics-based tasks learned before and is low when a new type of task emerges.

Second, social relationships during learning matter. As P2's narrative indicates and the narratives of the other participants suggest implicitly, there is much value in learning analytics skills and knowledge in a social setting. The value of social learning is seen in its capacity to provide learners with immediate feedback, help them validate ideas and insights obtained in learning processes, and offer additional guidance when necessary. This social dimension of learning becomes absent when professionals learn to perform specific tasks independently in accordance with instructions provided in videos (e.g., P5).

Third, organisational environment acts as a moderator, either facilitating or impeding individuals' learning and readiness. The difference between the participants like P2, P3, and P4, who were working for companies that were highly analytically oriented, and those

like P5, whose employers did not promote analytics, became apparent from the interviews. Individuals working in analytically-oriented companies were more advanced in terms of developing their analytics readiness. Given the moderating function played by organisational context, efforts targeted at individual-level changes would not be very productive unless organisational context was taken into account.

Fourth, the quality of educational background matters. Professionals who found their educational background to be adequate reported themselves to be readier than those who had doubts about the usefulness of their education. Their educational background turned out to play this positive role not because they had received a sufficient amount of analytics knowledge from professors but because the conceptual understanding acquired through formal education could later serve as a basis for developing analytics skills independently.

5.10 Reflections on the Study's Contribution to Practice

While the practical contribution of this study goes beyond the recommendations provided in Sections 6.4, they should not be discounted in evaluating its impact. Indeed, one way this research might be used by accounting practitioners is in helping them diagnose and address the gaps in their own analytics readiness. The operationalization of analytics readiness achieved by the survey items developed for this study, such as tool usage, assessed skill levels, perceptions of readiness, skill acquisition mechanisms, and organizational readiness provide useful indicators that organizations and professional bodies might choose to adopt when conducting their own diagnostics.

More specifically, the conditional readiness construct proposed in this study holds diagnostic value for practitioners. While it would not be sufficient to assess whether professionals are ready for analytics, the conditional readiness perspective prompts practitioners to ask themselves a number of important questions. Namely, under which conditions is readiness most vulnerable? For what kinds of tasks do accounting professionals feel ready? Where does their analytical capacity end? The answers to those questions will allow for more tailored approaches toward improving analytics skills and readiness than mere measurement and comparison of self-reported readiness scores.

Furthermore, this research is likely to be helpful to organizations and educators when designing and implementing analytics training programs. As shown by this research, readiness tends to be both task-specific and conditional, meaning that a good training program should not only increase aggregate self-reported readiness scores but also expand the variety of tasks for which accountants feel ready. Indeed, a program that achieves the former goal but does not achieve the latter represents wasted effort.

6. Conclusions

This chapter marks the concludes of this thesis as it summarises its findings, insights, and implications. The structure of this chapter follows the same order as the thesis outline presented in Section 1.7. First, this chapter presents findings in response to the four sub-research questions and the main research question. Then, theoretical contributions are identified before providing practical implications for three different parties. Afterwards, ethical implications, social implications, limitations of the study, and recommendations for future studies are discussed.

The research problem is stated as follows: 'To what extent are accounting professionals working in Nigeria equipped with data analytics skills required in a data driven accounting environment?' Using a convergent mixed methods design involving an online questionnaire administered to 125 accounting and finance professionals and six semi structured interviews, this study has developed a substantial amount of evidence that is reliable and valid. By analysing the evidence using role theory (Biddle, 1986), human capital theory (Becker, 1993), and professional competence theory (Mulder, 2014), this study found that accounting professionals in Nigeria are partially and conditionally prepared for the data driven era.

6.1 Summary of Main Findings

Findings from empirical research were consistent and converged in respect of all sub research questions.

Concerning tools usage and skills, advance Excel, including the use of pivoting table, Power Query and dashboards was the most commonly used analytics tool, with the frequency of its use being reported by 51.2 % of survey participants. In regard to other analytics tools, usage of Power BI and Tableau is quite common (high frequency 22.4 %), while 47.2 % report a low frequency. Six interviewees confirmed their analytical activity as applied and operational, focusing on improving the process of operation and not forecasting or strategic activities. As a result, the research findings suggest that even though analytics became part of professional practice, its development is still partially achieved, failing to correspond to expectations of both organizations and accounting profession bodies (ACCA and ICAN).

Concerning readiness, 61.6 % of the respondents claim feeling prepared enough to deal with analytical tasks, whereas 74.4 % feel confident working with data and making managerial decision based on it. However, the findings should be interpreted with caution as qualitative data revealed readiness to be task-specific and conditional ready for known and familiar analytical processes, but struggling with unfamiliar and new situations. It can be said that participants P3 and P6 are very clear about these conditions. Concerning predictors of readiness, chi square analysis suggests educational preparation (Cramer's V = 0.48), confidence using data in making management decisions (V = 0.44) and organizational support (V = 0.41) as the most influential factors. Surprisingly, years of experience, position in organization, and qualifications have no significant impact on readiness (years of experience p = .562, position in organization p = .260, and qualificationsp=.684).

Concerning the path of skill acquisition, 61.6% of participants gained their analytics skills mainly through self-taught and experiential methods, 25.6% relied on professional training, and 12.8% reported not having developed any analytics skills. Participants in interviews regardless of the career level mentioned YouTube tutorials, Udemy and CFI courses, and experimenting in workplaces as the main learning sources. Formal university education was viewed as theoretical and ineffective in practice. P1, P4, and P6 specifically pointed out the lack of practical hands-on experience in tools in university education. The statistically significant relationship between the path of skill acquisition and readiness (Cramer's V = 0.30) indicates the influence of the path of skill acquisition on the state of readiness.

Concerning challenges, there were common barriers among two strands: the lack of formal and structured training provision, difficulties in accessing software, time pressure due to workload, and losing skills due to infrequent use. P5 specifically highlighted the poor focus on developing analytics skills in Nigeria organizations as a challenge. Barriers are further aggravated by infrastructure issues related to expensive licensing fees and varying levels of investment in continuous professional development programs, which is supported by Fijabi and Lasisi (2023, p. 71-79).

In general, the results provide an integrative picture of the evolving industry. Indeed, there is clear evidence of the evolution of role expectations towards being based on analytical contributions, which is in accordance with role theory, which posits role change facilitated by institutions (Biddle, 1986; Burns and Baldvinsdottir, 2005). Professionals have already made some changes in response by adopting self-learning practices. However, institutional support has not been properly developed yet.

6.2 Theoretical Contributions

In summary, this study makes four key contributions to the professional literature on data analytics and the accounting profession.

First, this study advances the existing professional competence theory (Mulder, 2014) by operationalising and providing an empirical base to the concept of conditional readiness, defined above in Section 5.4 as the task-specific, practice-contingent and situationally fragile form of analytics competence. While previous studies based on this theory have considered readiness as a relatively stable feature of professionals, this study demonstrates the extent to which it depends on both the type of analysis performed and the presence of organisational support. Thus, the findings provide theoretical underpinning for future research aimed at measuring and developing analytics readiness.

Second, this study also contributes to the human capital theory (Becker, 1993) in demonstrating the significance of the informal and autonomous learning modes in building analytics human capital in an emerging economy context. The conclusion that neither years of experience nor the level of formal education can predict analytics readiness in a significant way questions the linear model of professional human capital development. Instead, this finding supports Nafukho et al.'s (2004, p. 545-546) argument that the gains from human capital investments depend on the relevancy of the acquired competencies to the contemporary professional demands and not the number of years invested in learning.

Third, this study contributes to role theory (Biddle, 1986) in demonstrating the existence of a role gap in Nigeria, between the analytics-related expectations set externally by institutions, and the actual analytics practices engaged in by professionals. The role ambiguity and role conflict identified by a few participants reflect the structural discrepancy that role theory predicts under external demands exceeding the institutional preparedness. By confirming and extending this phenomenon observed in developed economies earlier by Burns and Baldvinsdottir (2005) and Hopper (1980), this study adds to the knowledge base regarding its generalisability across national contexts.

Finally, by focusing on Nigeria, this dissertation provides empirical insight into the growing field of data analytics in emerging economies' accounting practices. As Bhimani and Willcocks (2014, p. 470-473) rightly noted, digital transformation in accounting has context-specific implications. This study highlights and illustrates some of the specific factors contributing to the current state of analytics readiness in Nigeria factors that cannot be accounted for when studying the same phenomenon in developed economy contexts.

6.3 Practical Recommendations

There are several implications derived from the results of this research that can be addressed through actions by different stakeholders.

Firstly, higher education institutions offering accounting programmes in Nigeria must conduct a comprehensive audit of their current educational programmes in relation to practical analytics skills development. The message from this study is clear P1, P4, and P6 all described their university experience as being theoretically sufficient yet not practically adequate in regard to the level of proficiency needed for analytics work. In other words, universities need to integrate practical training related to working with such tools as advanced Excel, Power Query, Power BI, and elementary statistical analysis techniques using case-based teaching as well as engaging partners from the professional field and employers in the process of developing a relevant curriculum.

Secondly, professional bodies, namely ICAN and ACCA, should include the element of analytics skill enhancement in their CPD framework. In view of the fact that self-learning is currently seen as a major driver of competence improvement, professional bodies are capable of assuming a facilitating role by providing quality learning materials for self-upskilling purposes and formalising the process through inclusion into CPD credits. Furthermore, it would make sense for these bodies to develop progressive levels of competency in analytics and introduce it gradually into their professional examinations thus guaranteeing that future professionals will possess sufficient practical knowledge.

Finally, employing organisations in Nigeria must understand that the issue of analytics readiness is heavily influenced by the organisational environment. The statistically significant correlation between organisational support and analytics readiness ($V = 0.41$) indicates that structured training programmes, mentorship, and access to proper tools bring tangible benefits in this regard. The difference in the experiences of participants in this study shows the effect of lack of organisational support clearly P2 and P4 mentioned organisations that provided opportunities for developing these skills whereas P5 indicated the absence of any such opportunities.

6.4 Ethical and Societal Implications

There are certain ethical and societal considerations that should be taken into account regarding the outcomes of the present study.

First of all, it could be noted that there exists an inherent risk associated with excessive reliance on the self-directed learning and financing provided by the organisation as two of the crucial factors that determine the level of readiness of accountants to analytics. Accountants who operate in poor working conditions and those organisations that cannot afford paying for employees' learning activities become victims of unfair treatment and face difficulties in attaining the needed level of skills required to advance their career further. If left uncontrolled, this trend will create a situation of segregation in the Nigerian accounting profession as the ability of accountants to perform data-driven tasks will depend on the nature of their employment. Such a situation will contradict professional norms and principles established by ICAN and ACCA.

From a wider perspective, the discussed results are relevant to the quality of financial information, audits, and accountability in Nigeria. It should be noted that accounting professionals have an essential responsibility of providing accurate financial information, which is then used for decision-making, regulatory, and accountability purposes (Iyoha and Oyerinde, 2010, p. 361-364). Thus, if the accounting profession in Nigeria lacks sufficient skills to handle data-based analysis, this fact may lead to a situation of misstated financial reports, which may negatively impact organisational decision-making and accountability.

Regarding ethical considerations related to this study, it can be stated that it was performed following the ethical requirements of Umeå University's School of Business, Economics, and Statistics. Consent from all participants was acquired before collecting data. Furthermore, all respondents are referred to in this paper by their anonymous code (P1-P6), and their personal details were never recorded or retained. The questionnaires were completed anonymously. Data collection procedures were approved by the supervising team. There was no external financing involved, nor any conflicts of interests were taken into consideration when designing and conducting the study. Concerning the use of artificial intelligence, the authors confirm that all activities in this respect were conducted in accordance with Umeå University's academic integrity policies outlined in Section 3.8.

6.5 Limitations of the Study

In summary, there are several aspects that should be considered when interpreting the outcomes of the presented research.

Firstly, self-reporting regarding the extent of conditional readiness and assessing the skill set of the employees may be subject to bias, as people tend to distort information under the influence of social desirability bias in order to appear in the best possible light. However, this problem has been partially mitigated by using the method of semi-structured interviews but the absence of objective evaluation tools still remains the problem.

Secondly, the use of non-probability purposive sampling reduces the possibility for generalization of the survey findings, which cannot be projected on the entire population of Nigerian accounting professionals. This sample is comprised primarily of young accountants with 3 to 5 years of work experience who are involved in financial services and manufacturing. As stated in section 3.7, it is recommended to view the findings in terms of analytical rather than statistical generalization.

Thirdly, although the convergent design increases the internal validity of this research, interpretive judgments have been used in order to resolve the discrepancy between the comparatively high survey scores and the more conditional qualitative findings. In particular, preference has been given to the latter, as the findings were more context-specific.

Fourthly, the cross-sectional design provides a snapshot of the readiness level at one point in time , early 2026. Although it can serve as a good measure of assessing the state of affairs, longitudinal studies would be needed for evaluating the development of capabilities in the future. However, the longitudinal design could not be employed in this thesis due to its complexity.

Finally, even though the sample of six respondents who took part in interviews has been chosen purposively, it could be increased by adding individuals from other industries and various career stages. Even though the saturation of the main themes was achieved, further insights might be drawn from additional participants.

6.6 Directions for Future Research

As for future research ideas suggested by the findings of this study, there are five productive directions worth exploring further.

First, the need to include an objective measure of skill development in future analyses of readiness is apparent from the results of this study. Conducting some kind of performance test measuring actual proficiency in certain analytics skills (such as data cleansing, dashboard creation, etc.) would provide important insights regarding whether the conditionally-readiness of accounting professionals comes from underdeveloped skills or lack of confidence in them.

The next point that could benefit further research, according to this study, is related to a need for greater elaboration on the role of organizational support as one of the key determinants of readiness. Further research could address which type(s) of organizational initiatives can help accountants develop their skills, such as mentorship, training programs, licensed software or networks of professional collaboration.

Next, the current study was conducted solely from the perspective of the professionals themselves. Including employers', mentors', and professional organizations' perspectives in future studies may be beneficial as it could reveal some differences in perception of one's readiness level between the professional themselves and other parties involved.

Fourth, a longitudinal approach, in which the cohort of accountants would be followed for a number of years, might help to discover how the development of their analytics readiness unfolds during different phases of their career path.

Lastly, comparing African countries and potentially other developing economies in terms of the described dynamics (dominance of informal learning, mismatches between educational background and job-related skills, differences in organizational support), would enrich the body of literature on the matter.

6.7 Closing Statement

The results of this study have yielded a richly empirical and deeply theoretical insight into data analytics readiness amongst accountants in Nigeria. It has been proven that the field is in a true state of transition where professional roles have become definitively more data driven and individuals within these roles demonstrate innovation and ingenuity as they respond to the changing environment. In addition, it has been demonstrated that analytical software tools are widely used by the respondents, and the desire to increase capability in the field is widespread.

Despite this progress, however, the study has identified a readiness deficit in terms of structure. Namely, such factors as insufficient practical training in formal education systems, unequal provision of organisational support, as well as infrastructural limitations in the environment of professional practice in Nigeria prevent Nigerian accounting professionals from becoming fully capable in applying their analytical skills effectively and consistently. Indeed, the notion of conditional readiness, discussed in Chapter 5 and elaborated on throughout this study, captures the essence of the problem. Accountants in Nigeria are ready and able to learn analytics, but what is needed for that learning to happen is the necessary infrastructure. This challenge needs to be overcome by concerted actions of all three institutions involved.

6.8 Contribution to the Business Administration Field

The present study contributes to Business Administration literature at the confluence of three areas of academic discourse: data analytics in accounting practice, professional competence and capability development, and business and professional practice in Africa and emerging economies. In terms of the accounting analytics literature, the study provides the first empirical analysis of analytics readiness among practicing accounting professionals in Nigeria, addressing the gap pointed out by Sutton et al. (2016, p. 68) and Appelbaum et al. (2017, p. 29-35).

In the professional competence literature, the concept of conditional readiness stands as a truly original theoretical contribution. Previous research related to professional competence has largely regarded readiness as a fairly stable individual characteristic, which can be measured via surveys and serves as a predictor of professional competence. The current study shows that the aggregate definition of readiness as proposed in previous literature is insufficient, and fails to capture significant empirical variation. The concept of conditional readiness offers a better description of how analytics readiness manifests in practice, and raises additional questions for future research: What conditions define boundaries of conditional readiness? How do professionals develop from conditional to general readiness? Under what conditions is greater generalisability of analytics competencies attained?

Finally, within the broader field of Business Administration focused on emerging economy professional practice, the study offers a theoretically informed empirical study

of one of the key challenges associated with digital transformation analytics readiness of accounting professionals set in the particular institutional and infrastructural environment of Nigeria. Beyond its specific focus on Nigeria, the study's analytical framework, which integrates role theory, human capital theory, and professional competence theory in a convergent mixed-methods design, could be used for studying similar phenomena in other emerging economy business and professional practices.

6.9 Policy Implications

The results of this study have relevance not only for professional bodies, educational institutions, and individual employers, but also for the policy environment under which accounting education and professional development takes place in Nigeria. There are several policy lessons to be derived from the data gathered.

First, there are implications for accounting education policy, in that the Nigerian Universities Commission and the relevant professional bodies should come together in order to set minimum requirements for the inclusion of practical analytics content in the curricula of accounting courses. The uniform conclusion made by participants from the various Nigerian Universities surveyed in the study that formal education did not equip candidates with the necessary skills indicates that this is a systemic issue that requires a systematic approach. Setting minimum requirements for the inclusion of relevant curriculum content, including the expectation that graduates will demonstrate competence in using at least two analytics tools, an understanding of statistics, and the ability to present and interpret their results would be one possible way forward.

Second, in terms of professional development policy, both ICAN and ACCA could examine the possibility of incorporating competences in practical analytics into their criteria for maintaining membership, along with specific CPD requirements for development in this area. As long as self-directed learning is relied upon as the only vehicle for developing these skills, it remains at the discretion of individual candidates, based on factors such as motivation, availability of resources, etc. By making such competences part of CPD criteria, a new set of institutional expectations could be established, and a more systematic approach towards building analytics competences within the profession could be developed.

Third, in terms of policy relating to employment and organisations, there may be a case for professional and regulatory bodies to explore how they can incentivise organisations operating in Nigeria to make investments in professional development equitably among their Nigerian workforce. The example provided by P5 of the international company that deliberately chose to invest in the development of their employees' analytics capabilities in other jurisdictions, but not in Nigeria itself demonstrates what amounts to professional discrimination in the context of analytics readiness, which leads to systemic consequences.

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Validity and Generalisability of the Theoretical Contribution

It is worth discussing the issue of the validity of conditional readiness as a theoretical contribution to the field. The construct has emerged through abduction based on the findings that were obtained both quantitatively and qualitatively, and the construct was formulated in the context of Nigerian accounting professionals as of 2026. Nevertheless, the question that needs to be answered at this point is whether conditional readiness can be considered as an inherent property of the development process of professional analytics competencies or this property is specific to the Nigerian context and the particular moment in history. There are several reasons why one should consider conditional readiness as a universal, not context-specific phenomenon.

First, the nature of the process that leads to acquiring skills in conditional readiness, i.e. self-directed, reactive, and urgent approach towards fulfilling the demands of the task, is universal regardless of the context in which such development is going on. Second, the pattern of conditional readiness that was found, namely conditional readiness for familiar tasks and weak readiness for unfamiliar tasks is also a predictable outcome according to any professional development theory in the case where professional development is informal. Finally, inconsistency between aggregate levels of conditional readiness measured in surveys and qualitative measures is likely to be methodologically determined.Nevertheless, it is possible that conditional readiness is more pronounced in the Nigerian context because there is no extensive training programme in place in contrast to the developed world economies in which organisational readiness for professional development makes this phenomenon less intense. Future research should investigate whether the pattern of conditional readiness holds true in other professional contexts as well, and in what way its intensity varies depending on the strength of the institutional support for professional development.

As far as the external validity of quantitative findings is concerned, it should be kept in mind that in the course of the sampling procedure, LinkedIn selection bias may have influenced the results. This means that the relationship that emerged on the basis of the chi-square analysis cannot be taken as a precise estimate of the relationship in the overall Nigerian accounting professionals' population. Therefore, it would be necessary to replicate the study and use alternative methods of sampling such as survey distribution in collaboration with relevant organisations like ICAN and ACCA.

Appendix A: Survey Instrument

Data Analytics Skills and Professional Readiness in the Accounting Profession

This survey is part of a Masters thesis in Accounting and examines data analytics skills and professional readiness among accounting professionals. The survey takes approximately 7–10 minutes to complete. Participation is voluntary and anonymous, and responses will be used solely for academic research purposes. Thank you for your contribution.

Section 1: Background Information

1. What is your current role?

- Financial Accountant
- Management Accountant
- Financial Analyst
- Auditor
- Finance Manager
- Other

2. Years of professional experience:

- 0–2
- 3–5
- 6–10
- 11+

3. Industry:

- Financial services
- Manufacturing
- Public sector
- Consulting
- Other

4. Highest qualification:

- Bachelor
- Master
- Professional certification (ACCA, ICAN, etc.)
- Other

5. Country of primary professional practice:

- Nigeria
- Other

Section 2: Data Analytics Tool Usage

6. How frequently do you use the following tools in your work?

Scale: Never Rarely Sometimes Often Very Frequently

- Advanced Excel functions (Pivot tables, Power Query, etc.)
- Power BI / Tableau / Dashboard tools
- ERP analytics features
- Statistical software (R, Python, SPSS)
- Data visualisation tools

Section 3: Perceived Data Analytics Skill Level

7. How would you rate your current skill level in the following data analytics tasks?

Scale: Very Low Low Moderate High Very High

- Data cleaning and preparation
- Data visualisation
- Interpreting large datasets
- Building dashboards
- Using data for strategic decision-making

Section 4: Perceived Readiness

8. How strongly do you agree with the following statements about analytics capability and support in your role?

Scale: Strongly Disagree Disagree Neutral Agree Strongly Agree

- I feel adequately prepared to perform analytics-related tasks in my role.
- My education prepared me well for data analytics demands.
- My organisation provides sufficient support for developing analytics skills.
- I am confident using data to support management decisions.
- Data analytics skills are essential for success in my current role.

Section 5: Skill Acquisition Pathways

9. Where did you primarily develop your data analytics skills?

- Formal university education
- Professional training programs
- On-the-job experience
- Self-learning (online courses, YouTube, etc.)
- I have not developed such skills

Section 6: Challenges

10. To what extent do you agree with the following statements about barriers to developing and using analytics skills in your role?

Scale: Strongly Disagree Disagree Neutral Agree Strongly Agree

- Lack of formal training limits my analytics capabilities.
- Limited access to software tools affects my ability to use analytics.
- Time constraints prevent me from developing analytics skills.
- I feel role pressure to perform analytics tasks beyond my training.

Appendix B: Interview Guide

Opening

Can you describe your current role and responsibilities?

Role Evolution

How has your role changed over the past few years?

Have data analytics responsibilities increased? In what way?

Skill and Readiness

How confident do you feel performing analytics-related tasks?

Can you give examples of situations where you used data analytics?

Skill Development

Where did you learn most of your analytics skills?

Did your university education prepare you adequately?

What role has self-learning played?

Organisational Context

Does your organisation support analytics skill development?

What barriers do you face when using analytics tools?

Closing

In your opinion, what skills will accountants need in the next 5–10 years?

Appendix C: Participant Information and Consent

All interview participants received a Participant Information Sheet explaining the study purpose, voluntary nature of participation, and confidentiality procedures. Informed consent was confirmed verbally at the start of each recorded interview. Survey participants provided consent via an introductory consent screen. All participants are identified solely by anonymous codes (P1–P6) throughout this thesis.